

- LOW VOLTAGE FIRE RESISTANT CABLES
- FLEXIBLE CABLES
- MEDIUM VOLTAGE HIGH VOLTAGE EXTRA HIGH VOLTAGE CABLES
- LOW VOLTAGE POWER & INSTRUMENTATION CABLES



Asian Wire and Cable

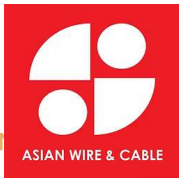
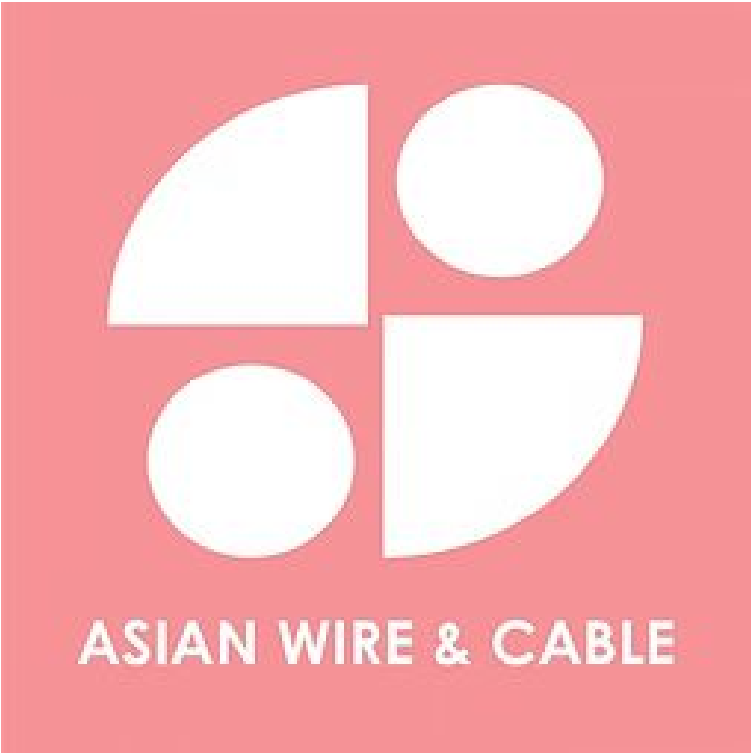
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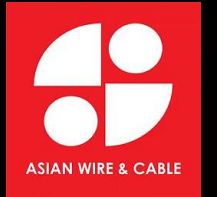
# ASIAN WIRE & CABLE

MID VOLTAGE  
HIGH VOLTAGE  
EXTRA HIGH VOLTAGE

Asian Wire and Cable





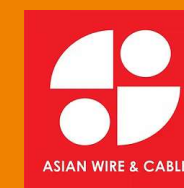


# UNPARALLED SUCCESS<sup>7</sup> SINCE 1993<sub>L</sub>

- LOW VOLTAGE FIRE RESISTANT CABLES
- FLEXIBLE CABLES
- ✓ MEDIUM VOLTAGE HIGH VOLTAGE EXTRA HIGH VOLTAGE CABLES
- LOW VOLTAGE POWER & INSTRUMENTATION CABLES



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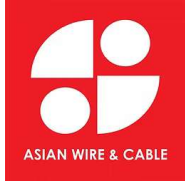


OUR VISION WORLD LEADER  
THAT SETS THE BECNHMARK FOR  
QUALITY AND EXCELLENCE

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# About Us



TEAMWORK



INNOVATION



WIN-WIN

Established in 1993, Asian Wire and Cable specializes in manufacturing wires and cables. The company is also an enterprise collection of R & D, Sales and Services. The main products include Power Cables, Electrical Cables, Control Cables, Housing Wires (CCC Approval), Mineral Insulation Cables, Prefabricated Branch Cables.

Our mission is to consistently engineer, produce, and deliver exceptional cable solutions that empower connectivity across industries and geographies. By adhering to the highest standards of innovation, craftsmanship, and environmental responsibility, we aim to exceed our customers' expectations and contribute to the advancement of global technological progress. Through collaborative partnerships, cutting-edge research, and a commitment to continuous improvement, we strive to create lasting value for our customers, employees, stakeholders, and the communities we serve. Our unwavering dedication to excellence drives us forward on the path to becoming the foremost global leader for superior cables that power the connections of today and tomorrow.

**WORLD-CLASS MANUFACTURING** We are committed to bring world-class manufacturing to our customers through standardization of processes and manufacturing high quality products. Striving to improve and standardize our Production models and business processes, we also aim to reduce lead time and manufacturing costs of building materials.

#### UNPARALLELED CUSTOMER SERVICE

In this highly competitive industry, we are also driven to focus on our R&D in order to fulfil an extensive range of our customers' requirements and standards.

**STRINGENT QUALITY CONTROL** We believe that our strong customers' satisfaction comes from our "Best or Nothing" quality control. We have a strong quality control system to ensure that we complete every project to the highest industry standards. We have a strong quality control system that begins from the purchasing, warehouse, production, packaging, and transportation to ensure professional quality is maintained throughout.

With this commitment, we are able to assist our customers to manufacture cables of various standards that include CCC, ISO9002, CE, ROHS, VDE, TIR, SAA, SON CAP, SIRIM, UL, CSA, JET, SAA, SEMKO, NEMKO, FIMKO, DEMKO, SEV, OVE, CEBEC, CNS, PSB, etc. At Asian Wire and Cable, we believe that with quality comes longevity and with customer satisfaction comes growth.



# Our Mission



Win-win



Innovation



Teamwork

Our mission is to consistently engineer, produce, and deliver exceptional cable solutions that empower connectivity across industries and geographies. By adhering to the highest standards of innovation, craftsmanship, and environmental responsibility, we aim to exceed our customers' expectations and contribute to the advancement of global technological progress. Through collaborative partnerships, cutting-edge research, and a commitment to continuous improvement, we strive to create lasting value for our customers, employees, stakeholders, and the communities we serve. Our unwavering dedication to excellence drives us forward on the path to becoming the foremost global leader for superior cables that power the connections of today and tomorrow.

## OUR BUSINESS PHILOSOPHY

WE VALUE RELATIONSHIPS &  
SUSTAIN TRUST WITH OUR  
VALUED STAKEHOLDERS





# Production Equipment



Good Service with Sincerity



# Production Equipment







# Quality Inspection Center



Good Service  
with Sincerity





# Product Details

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## MEDIUM VOLTAGE CABLES 1.8/3(3.6) kV, , NON-ARMOUR 1 CORE CU/XLPE/CTS/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                                                                                    |                                                                                                |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Conductor :                                                                        | Compact round stranded anneal copper                                                           |
| Conductor Screen :                                                                 | Semi-conducting Cross-linked Polyethylene                                                      |
| Insulation :                                                                       | XLPE (Cross-linked Polyethylene)                                                               |
| Insulation Screen :                                                                | Semi-conducting Cross-linked Polyethylene                                                      |
| Metallic Shield :                                                                  | Annealed copper tape (CTS)                                                                     |
| Filler :                                                                           | Non-hygroscopic material                                                                       |
| Wrapping Tape :<br>Inner Sheath :<br>Armour :<br>Wrapping Tape :<br>Outer Sheath : | Polyester / Non-woven tape - -<br>Polyester / Non-woven tape<br>Black PVC (Polyvinyl Chloride) |

### Technical Data:

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| Voltage :                               | 1.8/3(3.6) kV                                                     |
| Reference :                             | IEC 60228 & IEC 60502-1                                           |
| AC Testing<br>Voltage :                 | 6.5kV                                                             |
| Max. Conductor<br>Temperature :         | 90oC (normal operation)                                           |
| Max. Conductor<br>Temperature :         | 250oC (short circuit at 5s max.<br>duration)                      |
| Special<br>Properties upon<br>request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor Core x (mm <sup>2</sup> ) | Number and diameter of wires (no/mm) | Thickness of insulation (mm) | Nom. Dia. Of Armour Wire (mm) | Thickness of oversheath (mm) | Approx overall diameter (mm) | Approx net weight (kg/km) | Maximum Conductor Resistance at 20oC (Ohm/km) | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|---------------------------------------------------------------------|--------------------------------------|------------------------------|-------------------------------|------------------------------|------------------------------|---------------------------|-----------------------------------------------|------------------------------------------------|----------------------------------------------|
| 1 x 10                                                              | 6                                    | 2.0                          | -                             | 1.4                          | 14                           | 280                       | 1.83                                          | 90                                             | 100                                          |
| 1 x 16                                                              | 6                                    | 2.0                          | -                             | 1.4                          | 15                           | 360                       | 1.15                                          | 115                                            | 125                                          |
| 1 x 25                                                              | 6                                    | 2.0                          | -                             | 1.5                          | 17                           | 470                       | 0.727                                         | 150                                            | 165                                          |
| 1 x 35                                                              | 6                                    | 2.0                          | -                             | 1.5                          | 18                           | 590                       | 0.524                                         | 180                                            | 200                                          |
| 1 x 50                                                              | 6                                    | 2.0                          | -                             | 1.5                          | 19                           | 750                       | 0.387                                         | 210                                            | 240                                          |
| 1 x 70                                                              | 12                                   | 2.0                          | -                             | 1.6                          | 21                           | 970                       | 0.268                                         | 260                                            | 295                                          |
| 1 x 95                                                              | 15                                   | 2.0                          | -                             | 1.7                          | 23                           | 1250                      | 0.193                                         | 310                                            | 365                                          |
| 1 x 120                                                             | 18                                   | 2.0                          | -                             | 1.7                          | 24                           | 1510                      | 0.153                                         | 350                                            | 420                                          |
| 1 x 150                                                             | 18                                   | 2.0                          | -                             | 1.8                          | 26                           | 1820                      | 0.124                                         | 395                                            | 475                                          |
| 1 x 185                                                             | 30                                   | 2.0                          | -                             | 1.8                          | 28                           | 2180                      | 0.0991                                        | 450                                            | 545                                          |
| 1 x 240                                                             | 34                                   | 2.0                          | -                             | 1.9                          | 30                           | 2750                      | 0.0754                                        | 520                                            | 640                                          |
| 1 x 300                                                             | 34                                   | 2.0                          | -                             | 2.0                          | 33                           | 3360                      | 0.0601                                        | 585                                            | 740                                          |
| 1 x 400                                                             | 53                                   | 2.0                          | -                             | 2.1                          | 36                           | 4370                      | 0.0470                                        | 670                                            | 850                                          |



## MEDIUM VOLTAGE CABLES 3.6/6(7.2) kV, XLPE INSULATED, PVC SHEATHED NON-ARMOUR 1 CORE CU/XLPE/CTS/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape - -            |
| Inner Sheath :      | Polyester / Non-woven tape                |
| Armour :            | Black PVC (Polyvinyl Chloride)            |
| Wrapping Tape :     |                                           |
| Outer Sheath :      |                                           |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 3.6/6(7.2) kV                                                     |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 12.5kV                                                            |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor<br>Core x (mm <sup>2</sup> ) | Number and diameter of wires<br>(no/mm) | Thickness of insulation<br>(mm) | Nom. Dia. Of Armour Wire<br>(mm) | Thickness of oversheath<br>(mm) | Approx overall diameter<br>(mm) | Approx net weight<br>(kg/km) | Maximum Conductor Resistance at 20oC<br>(Ohm/km) | Ampacities direct burial in ground at 30oC<br>(A) | Ampacities in free air at 40oC (ambient)<br>(A) |
|------------------------------------------------------------------------|-----------------------------------------|---------------------------------|----------------------------------|---------------------------------|---------------------------------|------------------------------|--------------------------------------------------|---------------------------------------------------|-------------------------------------------------|
| 1 x 10                                                                 | 6                                       | 2.5                             | -                                | 1.4                             | 16                              | 310                          | 1.83                                             | 90                                                | 100                                             |
| 1 x 16                                                                 | 6                                       | 2.5                             | -                                | 1.5                             | 17                              | 400                          | 1.15                                             | 115                                               | 125                                             |
| 1 x 25                                                                 | 6                                       | 2.5                             | -                                | 1.5                             | 18                              | 510                          | 0.727                                            | 150                                               | 165                                             |
| 1 x 35                                                                 | 6                                       | 2.5                             | -                                | 1.5                             | 19                              | 620                          | 0.524                                            | 180                                               | 200                                             |
| 1 x 50                                                                 | 6                                       | 2.5                             | -                                | 1.6                             | 21                              | 800                          | 0.387                                            | 210                                               | 240                                             |
| 1 x 70                                                                 | 12                                      | 2.5                             | -                                | 1.6                             | 22                              | 1010                         | 0.268                                            | 260                                               | 295                                             |
| 1 x 95                                                                 | 15                                      | 2.5                             | -                                | 1.7                             | 24                              | 1290                         | 0.193                                            | 310                                               | 365                                             |
| 1 x 120                                                                | 18                                      | 2.5                             | -                                | 1.7                             | 25                              | 1550                         | 0.153                                            | 350                                               | 420                                             |
| 1 x 150                                                                | 18                                      | 2.5                             | -                                | 1.8                             | 27                              | 1870                         | 0.124                                            | 395                                               | 475                                             |
| 1 x 185                                                                | 30                                      | 2.5                             | -                                | 1.9                             | 29                              | 2240                         | 0.0991                                           | 450                                               | 545                                             |
| 1 x 240                                                                | 34                                      | 2.6                             | -                                | 2.0                             | 32                              | 2830                         | 0.0754                                           | 520                                               | 640                                             |
| 1 x 300                                                                | 34                                      | 2.8                             | -                                | 2.0                             | 35                              | 3450                         | 0.0601                                           | 585                                               | 740                                             |
| 1 x 400                                                                | 53                                      | 3.0                             | -                                | 2.2                             | 38                              | 4510                         | 0.0470                                           | 670                                               | 850                                             |

## MEDIUM VOLTAGE CABLES 8.7/15(17.5) kV, XLPE INSULATED, PVC SHEATHED NON-ARMOUR 1 CORE CU/XLPE/CTS/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape - -            |
| Inner Sheath :      | Polyester / Non-woven tape                |
| Armour :            | Black PVC (Polyvinyl Chloride)            |
| Wrapping Tape :     |                                           |
| Outer Sheath :      |                                           |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 8.7/15(17.5) kV                                                   |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 30.5kV                                                            |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor<br>Core x (mm <sup>2</sup> ) | Number and diameter of wires<br>(no/mm) | Thickness of insulation<br>(mm) | Nom. Dia. Of Armour Wire<br>(mm) | Thickness of oversheath<br>(mm) | Approx overall diameter<br>(mm) | Approx net weight<br>(kg/km) | Maximum Conductor Resistance at 20oC<br>(Ohm/km) | Ampacities direct burial in ground at 30oC<br>(A) | Ampacities in free air at 40oC (ambient)<br>(A) |
|------------------------------------------------------------------------|-----------------------------------------|---------------------------------|----------------------------------|---------------------------------|---------------------------------|------------------------------|--------------------------------------------------|---------------------------------------------------|-------------------------------------------------|
| 1 x 10                                                                 | -                                       | -                               | -                                | -                               | -                               | -                            | -                                                | -                                                 | -                                               |
| 1 x 16                                                                 | -                                       | -                               | -                                | -                               | -                               | -                            | -                                                | -                                                 | -                                               |
| 1 x 25                                                                 | 6                                       | 4.5                             | -                                | 1.6                             | 22                              | 660                          | 0.727                                            | 140                                               | 155                                             |
| 1 x 35                                                                 | 6                                       | 4.5                             | -                                | 1.7                             | 24                              | 790                          | 0.524                                            | 170                                               | 190                                             |
| 1 x 50                                                                 | 6                                       | 4.5                             | -                                | 1.7                             | 25                              | 970                          | 0.387                                            | 200                                               | 230                                             |
| 1 x 70                                                                 | 12                                      | 4.5                             | -                                | 1.8                             | 27                              | 1210                         | 0.268                                            | 250                                               | 285                                             |
| 1 x 95                                                                 | 15                                      | 4.5                             | -                                | 1.8                             | 28                              | 1490                         | 0.193                                            | 295                                               | 355                                             |
| 1 x 120                                                                | 18                                      | 4.5                             | -                                | 1.9                             | 30                              | 1770                         | 0.153                                            | 335                                               | 410                                             |
| 1 x 150                                                                | 18                                      | 4.5                             | -                                | 1.9                             | 31                              | 2090                         | 0.124                                            | 375                                               | 455                                             |
| 1 x 185                                                                | 30                                      | 4.5                             | -                                | 2.0                             | 33                              | 2470                         | 0.0991                                           | 430                                               | 525                                             |
| 1 x 240                                                                | 34                                      | 4.5                             | -                                | 2.1                             | 36                              | 3060                         | 0.0754                                           | 495                                               | 620                                             |
| 1 x 300                                                                | 34                                      | 4.5                             | -                                | 2.2                             | 38                              | 3700                         | 0.0601                                           | 560                                               | 720                                             |
| 1 x 400                                                                | 53                                      | 4.5                             | -                                | 2.3                             | 42                              | 4730                         | 0.0470                                           | 640                                               | 825                                             |



## MEDIUM VOLTAGE CABLES 12/20(24) kV, XLPE INSULATED, PVC SHEATHED NON-ARMOUR 1 CORE CU/XLPE/CTS/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

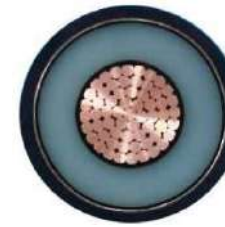
|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape - -            |
| Inner Sheath :      | Polyester / Non-woven tape                |
| Armour :            | Black PVC (Polyvinyl Chloride)            |
| Wrapping Tape :     |                                           |
| Outer Sheath :      |                                           |

### Technical Data:

|                                   |                                           |
|-----------------------------------|-------------------------------------------|
| Voltage :                         | 12/20(24) kV                              |
| Reference :                       | IEC 60228 & IEC 60502-2                   |
| AC Testing                        | 42kV                                      |
| Voltage :                         | 90oC (normal operation)                   |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration) |
| Max. Conductor Temperature :      | Flame-retardant sheath                    |
| Special Properties upon request : | LSHF sheath (PE) Polyethylene sheath      |

| Nominal cross-sectional area of conductor Core x (mm <sup>2</sup> ) | Number and diameter of wires (no/mm) | Thickness of insulation (mm) | Nom. Dia. Of Armour Wire (mm) | Thickness of oversheath (mm) | Approx overall diameter (mm) | Approx net weight (kg/km) | Maximum Conductor Resistance at 20oC (Ohm/km) | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|---------------------------------------------------------------------|--------------------------------------|------------------------------|-------------------------------|------------------------------|------------------------------|---------------------------|-----------------------------------------------|------------------------------------------------|----------------------------------------------|
| 1 x 10                                                              | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 16                                                              | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 25                                                              | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 35                                                              | 6                                    | 5.5                          | -                             | 1.8                          | 26                           | 890                       | 0.524                                         | 170                                            | 200                                          |
| 1 x 50                                                              | 6                                    | 5.5                          | -                             | 1.8                          | 27                           | 1070                      | 0.387                                         | 200                                            | 240                                          |
| 1 x 70                                                              | 12                                   | 5.5                          | -                             | 1.9                          | 29                           | 1320                      | 0.268                                         | 250                                            | 295                                          |
| 1 x 95                                                              | 15                                   | 5.5                          | -                             | 1.9                          | 31                           | 1600                      | 0.193                                         | 295                                            | 365                                          |
| 1 x 120                                                             | 18                                   | 5.5                          | -                             | 2.0                          | 32                           | 1890                      | 0.153                                         | 335                                            | 420                                          |
| 1 x 150                                                             | 18                                   | 5.5                          | -                             | 2.0                          | 34                           | 2210                      | 0.124                                         | 375                                            | 475                                          |
| 1 x 185                                                             | 30                                   | 5.5                          | -                             | 2.1                          | 36                           | 2600                      | 0.0991                                        | 430                                            | 545                                          |
| 1 x 240                                                             | 34                                   | 5.5                          | -                             | 2.2                          | 38                           | 3210                      | 0.0754                                        | 495                                            | 640                                          |
| 1 x 300                                                             | 34                                   | 5.5                          | -                             | 2.2                          | 41                           | 3830                      | 0.0601                                        | 560                                            | 740                                          |
| 1 x 400                                                             | 53                                   | 5.5                          | -                             | 2.3                          | 44                           | 4880                      | 0.0470                                        | 640                                            | 850                                          |

## MEDIUM VOLTAGE CABLES 18/30(36) kV, XLPE INSULATED, PVC SHEATHED NON-ARMOUR 1 CORE CU/XLPE/CTS/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape - -            |
| Inner Sheath :      | Polyester / Non-woven tape                |
| Armour :            | Black PVC (Polyvinyl Chloride)            |
| Wrapping Tape :     |                                           |
| Outer Sheath :      |                                           |

### Technical Data:

|                                   |                                           |
|-----------------------------------|-------------------------------------------|
| Voltage :                         | 18/30(36) kV                              |
| Reference :                       | IEC 60228 & IEC 60502-2                   |
| AC Testing                        | 63kV                                      |
| Voltage :                         | 90oC (normal operation)                   |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration) |
| Max. Conductor Temperature :      | Flame-retardant sheath                    |
| Special Properties upon request : | LSHF sheath (PE) Polyethylene sheath      |

| Nominal cross-sectional area of conductor Core x (mm <sup>2</sup> ) | Number and diameter of wires (no/mm) | Thickness of insulation (mm) | Nom. Dia. Of Armour Wire (mm) | Thickness of oversheath (mm) | Approx overall diameter (mm) | Approx net weight (kg/km) | Maximum Conductor Resistance at 20oC (Ohm/km) | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|---------------------------------------------------------------------|--------------------------------------|------------------------------|-------------------------------|------------------------------|------------------------------|---------------------------|-----------------------------------------------|------------------------------------------------|----------------------------------------------|
| 1 x 10                                                              | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 16                                                              | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 25                                                              | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 35                                                              | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 50                                                              | 6                                    | 8.0                          | -                             | 2.1                          | 33                           | 1360                      | 0.387                                         | 200                                            | 235                                          |
| 1 x 70                                                              | 12                                   | 8.0                          | -                             | 2.1                          | 34                           | 1600                      | 0.268                                         | 250                                            | 290                                          |
| 1 x 95                                                              | 15                                   | 8.0                          | -                             | 2.2                          | 36                           | 1910                      | 0.193                                         | 295                                            | 360                                          |
| 1 x 120                                                             | 18                                   | 8.0                          | -                             | 2.3                          | 38                           | 2220                      | 0.153                                         | 335                                            | 415                                          |
| 1 x 150                                                             | 18                                   | 8.0                          | -                             | 2.3                          | 39                           | 2550                      | 0.124                                         | 375                                            | 470                                          |
| 1 x 185                                                             | 30                                   | 8.0                          | -                             | 2.4                          | 41                           | 2960                      | 0.0991                                        | 430                                            | 540                                          |
| 1 x 240                                                             | 34                                   | 8.0                          | -                             | 2.4                          | 44                           | 3570                      | 0.0754                                        | 495                                            | 635                                          |
| 1 x 300                                                             | 34                                   | 8.0                          | -                             | 2.5                          | 46                           | 4230                      | 0.0601                                        | 560                                            | 735                                          |
| 1 x 400                                                             | 53                                   | 8.0                          | -                             | 2.6                          | 49                           | 5300                      | 0.047                                         | 640                                            | 845                                          |



MEDIUM VOLTAGE CABLES  
1.8/3(3.6) kV, XLPE INSULATED,  
PVC SHEATHED NON-ARMOUR  
3 CORES CU/XLPE/CTS/PVC



Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

Construction:

|                                                                                    |                                                                                                |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Conductor :                                                                        | Compact round stranded anneal copper                                                           |
| Conductor Screen :                                                                 | Semi-conducting Cross-linked Polyethylene                                                      |
| Insulation :                                                                       | XLPE (Cross-linked Polyethylene)                                                               |
| Insulation Screen :                                                                | Semi-conducting Cross-linked Polyethylene                                                      |
| Metallic Shield :                                                                  | Annealed copper tape (CTS)                                                                     |
| Filler :                                                                           | Non-hygroscopic material                                                                       |
| Wrapping Tape :<br>Inner Sheath :<br>Armour :<br>Wrapping Tape :<br>Outer Sheath : | Polyester / Non-woven tape - -<br>Polyester / Non-woven tape<br>Black PVC (Polyvinyl Chloride) |

Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 1.8/3(3.6) kV                                                     |
| Reference :                       | IEC 60228 & IEC 60502-1                                           |
| AC Testing Voltage :              | 6.5kV                                                             |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                |                                              |
| 3 x 10                                    | 6                            | 2.0                     | -                        | 1.8                     | 27                      | 900               | 1.83                                 | 85                                             | 80                                           |
| 3 x 16                                    | 6                            | 2.0                     | -                        | 1.9                     | 30                      | 1150              | 1.15                                 | 105                                            | 105                                          |
| 3 x 25                                    | 6                            | 2.0                     | -                        | 2.0                     | 33                      | 1510              | 0.727                                | 135                                            | 140                                          |
| 3 x 35                                    | 6                            | 2.0                     | -                        | 2.1                     | 36                      | 1880              | 0.524                                | 165                                            | 170                                          |
| 3 x 50                                    | 6                            | 2.0                     | -                        | 2.2                     | 39                      | 2430              | 0.387                                | 195                                            | 205                                          |
| 3 x 70                                    | 12                           | 2.0                     | -                        | 2.3                     | 43                      | 3120              | 0.268                                | 235                                            | 250                                          |
| 3 x 95                                    | 15                           | 2.0                     | -                        | 2.4                     | 47                      | 3990              | 0.193                                | 280                                            | 305                                          |
| 3 x 120                                   | 18                           | 2.0                     | -                        | 2.5                     | 50                      | 4840              | 0.153                                | 315                                            | 345                                          |
| 3 x 150                                   | 18                           | 2.0                     | -                        | 2.6                     | 53                      | 5820              | 0.124                                | 350                                            | 390                                          |
| 3 x 185                                   | 30                           | 2.0                     | -                        | 2.7                     | 57                      | 6920              | 0.0991                               | 395                                            | 445                                          |
| 3 x 240                                   | 34                           | 2.0                     | -                        | 2.9                     | 63                      | 8800              | 0.0754                               | 450                                            | 520                                          |
| 3 x 300                                   | 34                           | 2.0                     | -                        | 3.1                     | 68                      | 10790             | 0.0601                               | 495                                            | 580                                          |
| 3 x 400                                   | 53                           | 2.0                     | -                        | 3.3                     | 75                      | 14000             | 0.0470                               | 545                                            | 655                                          |

MEDIUM VOLTAGE CABLES  
3.6/6(7.2) kV, XLPE INSULATED,  
PVC SHEATHED NON-ARMOUR  
3 CORES CU/XLPE/CTS/PVC



Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

Construction:

|                                                                                    |                                                                                                |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Conductor :                                                                        | Compact round stranded anneal copper                                                           |
| Conductor Screen :                                                                 | Semi-conducting Cross-linked Polyethylene                                                      |
| Insulation :                                                                       | XLPE (Cross-linked Polyethylene)                                                               |
| Insulation Screen :                                                                | Semi-conducting Cross-linked Polyethylene                                                      |
| Metallic Shield :                                                                  | Annealed copper tape (CTS)                                                                     |
| Filler :                                                                           | Non-hygroscopic material                                                                       |
| Wrapping Tape :<br>Inner Sheath :<br>Armour :<br>Wrapping Tape :<br>Outer Sheath : | Polyester / Non-woven tape - -<br>Polyester / Non-woven tape<br>Black PVC (Polyvinyl Chloride) |

Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 3.6/6(7.2) kV                                                     |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 12.5kV                                                            |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                |                                              |
| 3 x 10                                    | 6                            | 2.5                     | -                        | 2.0                     | 32                      | 1010              | 1.83                                 | 85                                             | 80                                           |
| 3 x 16                                    | 6                            | 2.5                     | -                        | 2.0                     | 33                      | 1270              | 1.15                                 | 105                                            | 105                                          |
| 3 x 25                                    | 6                            | 2.5                     | -                        | 2.1                     | 36                      | 1640              | 0.727                                | 135                                            | 140                                          |
| 3 x 35                                    | 6                            | 2.5                     | -                        | 2.1                     | 38                      | 2000              | 0.524                                | 165                                            | 170                                          |
| 3 x 50                                    | 6                            | 2.5                     | -                        | 2.2                     | 42                      | 2560              | 0.387                                | 195                                            | 205                                          |
| 3 x 70                                    | 12                           | 2.5                     | -                        | 2.3                     | 45                      | 3260              | 0.268                                | 235                                            | 250                                          |
| 3 x 95                                    | 15                           | 2.5                     | -                        | 2.5                     | 49                      | 4160              | 0.193                                | 280                                            | 305                                          |
| 3 x 120                                   | 18                           | 2.5                     | -                        | 2.6                     | 52                      | 5020              | 0.153                                | 315                                            | 345                                          |
| 3 x 150                                   | 18                           | 2.5                     | -                        | 2.7                     | 56                      | 6010              | 0.124                                | 350                                            | 390                                          |
| 3 x 185                                   | 30                           | 2.5                     | -                        | 2.8                     | 59                      | 7180              | 0.0991                               | 395                                            | 445                                          |
| 3 x 240                                   | 34                           | 2.6                     | -                        | 3.0                     | 66                      | 9100              | 0.0754                               | 450                                            | 520                                          |
| 3 x 300                                   | 34                           | 2.8                     | -                        | 3.2                     | 72                      | 11160             | 0.0601                               | 495                                            | 580                                          |
| 3 x 400                                   | 53                           | 3.0                     | -                        | 3.4                     | 85                      | 14000             | 0.0470                               | 545                                            | 655                                          |



## MEDIUM VOLTAGE CABLES 6/10(12) kV, XLPE INSULATED, PVC SHEATHED NON-ARMOUR 3 CORES CU/XLPE/CTS/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape - -            |
| Inner Sheath :      | Polyester / Non-woven tape                |
| Armour :            | Black PVC (Polyvinyl Chloride)            |
| Wrapping Tape :     |                                           |
| Outer Sheath :      |                                           |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 6/10(12) kV                                                       |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 21kV                                                              |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                |                                              |
| 3 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 16                                    | 6                            | 3.4                     | -                        | 2.1                     | 38                      | 1490              | 1.15                                 | 105                                            | 105                                          |
| 3 x 25                                    | 6                            | 3.4                     | -                        | 2.2                     | 40                      | 1870              | 0.727                                | 135                                            | 140                                          |
| 3 x 35                                    | 6                            | 3.4                     | -                        | 2.3                     | 43                      | 2270              | 0.524                                | 165                                            | 170                                          |
| 3 x 50                                    | 6                            | 3.4                     | -                        | 2.4                     | 46                      | 2850              | 0.387                                | 195                                            | 210                                          |
| 3 x 70                                    | 12                           | 3.4                     | -                        | 2.5                     | 49                      | 3570              | 0.268                                | 235                                            | 255                                          |
| 3 x 95                                    | 15                           | 3.4                     | -                        | 2.6                     | 53                      | 4470              | 0.193                                | 280                                            | 310                                          |
| 3 x 120                                   | 18                           | 3.4                     | -                        | 2.7                     | 57                      | 5360              | 0.153                                | 315                                            | 350                                          |
| 3 x 150                                   | 18                           | 3.4                     | -                        | 2.8                     | 60                      | 6360              | 0.124                                | 350                                            | 395                                          |
| 3 x 185                                   | 30                           | 3.4                     | -                        | 3.0                     | 64                      | 7590              | 0.0991                               | 395                                            | 450                                          |
| 3 x 240                                   | 34                           | 3.4                     | -                        | 3.2                     | 70                      | 9480              | 0.0754                               | 450                                            | 525                                          |
| 3 x 300                                   | 34                           | 3.4                     | -                        | 3.3                     | 75                      | 11450             | 0.0601                               | 495                                            | 585                                          |
| 3 x 400                                   | 53                           | 3.4                     | -                        | 3.6                     | 82                      | 14760             | 0.0470                               | 545                                            | 660                                          |

## MEDIUM VOLTAGE CABLES 8.7/15(17.5) kV, XLPE INSULATED, PVC SHEATHED NON-ARMOUR 3 CORES CU/XLPE/CTS/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape - -            |
| Inner Sheath :      | Polyester / Non-woven tape                |
| Armour :            | Black PVC (Polyvinyl Chloride)            |
| Wrapping Tape :     |                                           |
| Outer Sheath :      |                                           |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 8.7/15(17.5) kV                                                   |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 30.5kV                                                            |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                |                                              |
| 3 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 25                                    | 6                            | 4.5                     | -                        | 2.4                     | 46                      | 2210              | 0.727                                | 135                                            | 140                                          |
| 3 x 35                                    | 6                            | 4.5                     | -                        | 2.5                     | 48                      | 2630              | 0.524                                | 165                                            | 170                                          |
| 3 x 50                                    | 6                            | 4.5                     | -                        | 2.6                     | 52                      | 3230              | 0.387                                | 195                                            | 210                                          |
| 3 x 70                                    | 12                           | 4.5                     | -                        | 2.7                     | 55                      | 3980              | 0.268                                | 235                                            | 255                                          |
| 3 x 95                                    | 15                           | 4.5                     | -                        | 2.8                     | 59                      | 4900              | 0.193                                | 280                                            | 310                                          |
| 3 x 120                                   | 18                           | 4.5                     | -                        | 2.9                     | 62                      | 5810              | 0.153                                | 315                                            | 350                                          |
| 3 x 150                                   | 18                           | 4.5                     | -                        | 3.0                     | 65                      | 6840              | 0.124                                | 350                                            | 395                                          |
| 3 x 185                                   | 30                           | 4.5                     | -                        | 3.1                     | 70                      | 8070              | 0.0991                               | 390                                            | 450                                          |
| 3 x 240                                   | 34                           | 4.5                     | -                        | 3.3                     | 76                      | 10000             | 0.0754                               | 445                                            | 525                                          |
| 3 x 300                                   | 34                           | 4.5                     | -                        | 3.5                     | 81                      | 1240              | 0.0601                               | 490                                            | 585                                          |
| 3 x 400                                   | 53                           | 4.5                     | -                        | 3.7                     | 88                      | 15370             | 0.0470                               | 540                                            | 660                                          |



MEDIUM VOLTAGE CABLES  
12/20(24) kV, XLPE INSULATED,  
PVC SHEATHED NON-ARMOUR  
3 CORES CU/XLPE/CTS/PVC



**Application**

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

**Construction:**

|                                                                                    |                                                                                                |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Conductor :                                                                        | Compact round stranded anneal copper                                                           |
| Conductor Screen :                                                                 | Semi-conducting Cross-linked Polyethylene                                                      |
| Insulation :                                                                       | XLPE (Cross-linked Polyethylene)                                                               |
| Insulation Screen :                                                                | Semi-conducting Cross-linked Polyethylene                                                      |
| Metallic Shield :                                                                  | Annealed copper tape (CTS)                                                                     |
| Filler :                                                                           | Non-hygroscopic material                                                                       |
| Wrapping Tape :<br>Inner Sheath :<br>Armour :<br>Wrapping Tape :<br>Outer Sheath : | Polyester / Non-woven tape - -<br>Polyester / Non-woven tape<br>Black PVC (Polyvinyl Chloride) |

**Technical Data:**

|                                   |                                           |
|-----------------------------------|-------------------------------------------|
| Voltage :                         | 12/20(24) kV                              |
| Reference :                       | IEC 60228 & IEC 60502-2                   |
| AC Testing                        | 42kV                                      |
| Voltage :                         | 90oC (normal operation)                   |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration) |
| Max. Conductor Temperature :      | Flame-retardant sheath                    |
| Special Properties upon request : | LSHF sheath (PE) Polyethylene sheath      |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC | Ampacities in free air at 40oC (ambient) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|--------------------------------------------|------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                        | (A)                                      |
| 3 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                          | -                                        |
| 3 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                          | -                                        |
| 3 x 25                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                          | -                                        |
| 3 x 35                                    | 6                            | 5.5                     | -                        | 2.6                     | 53                      | 2970              | 0.524                                | 165                                        | 175                                      |
| 3 x 50                                    | 6                            | 5.5                     | -                        | 2.7                     | 56                      | 3590              | 0.387                                | 195                                        | 215                                      |
| 3 x 70                                    | 12                           | 5.5                     | -                        | 2.8                     | 60                      | 4350              | 0.268                                | 235                                        | 260                                      |
| 3 x 95                                    | 15                           | 5.5                     | -                        | 3.0                     | 64                      | 5340              | 0.193                                | 280                                        | 315                                      |
| 3 x 120                                   | 18                           | 5.5                     | -                        | 3.1                     | 67                      | 6270              | 0.153                                | 315                                        | 355                                      |
| 3 x 150                                   | 18                           | 5.5                     | -                        | 3.2                     | 71                      | 7320              | 0.124                                | 350                                        | 400                                      |
| 3 x 185                                   | 30                           | 5.5                     | -                        | 3.3                     | 75                      | 8570              | 0.0991                               | 390                                        | 455                                      |
| 3 x 240                                   | 34                           | 5.5                     | -                        | 3.5                     | 81                      | 10540             | 0.0754                               | 445                                        | 525                                      |
| 3 x 300                                   | 34                           | 5.5                     | -                        | 3.7                     | 86                      | 12620             | 0.0601                               | 490                                        | 585                                      |
| 3 x 400                                   | 53                           | 5.5                     | -                        | 3.9                     | 93                      | 15990             | 0.0470                               | 540                                        | 660                                      |

MEDIUM VOLTAGE CABLES  
18/30(36) kV, XLPE INSULATED,  
PVC SHEATHED NON-ARMOUR  
3 CORES CU/XLPE/CTS/PVC



**Application**

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

**Construction:**

|                                                                                    |                                                                                                |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Conductor :                                                                        | Compact round stranded anneal copper                                                           |
| Conductor Screen :                                                                 | Semi-conducting Cross-linked Polyethylene                                                      |
| Insulation :                                                                       | XLPE (Cross-linked Polyethylene)                                                               |
| Insulation Screen :                                                                | Semi-conducting Cross-linked Polyethylene                                                      |
| Metallic Shield :                                                                  | Annealed copper tape (CTS)                                                                     |
| Filler :                                                                           | Non-hygroscopic material                                                                       |
| Wrapping Tape :<br>Inner Sheath :<br>Armour :<br>Wrapping Tape :<br>Outer Sheath : | Polyester / Non-woven tape - -<br>Polyester / Non-woven tape<br>Black PVC (Polyvinyl Chloride) |

**Technical Data:**

|                                   |                                           |
|-----------------------------------|-------------------------------------------|
| Voltage :                         | 18/30(36) kV                              |
| Reference :                       | IEC 60228 & IEC 60502-2                   |
| AC Testing                        | 63kV                                      |
| Voltage :                         | 90oC (normal operation)                   |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration) |
| Max. Conductor Temperature :      | Flame-retardant sheath                    |
| Special Properties upon request : | LSHF sheath (PE) Polyethylene sheath      |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC | Ampacities in free air at 40oC (ambient) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|--------------------------------------------|------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                        | (A)                                      |
| 3 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                          | -                                        |
| 3 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                          | -                                        |
| 3 x 25                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                          | -                                        |
| 3 x 35                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                          | -                                        |
| 3 x 50                                    | 6                            | 8.0                     | -                        | 3.1                     | 68                      | 4660              | 0.387                                | 195                                        | 215                                      |
| 3 x 70                                    | 12                           | 8.0                     | -                        | 3.2                     | 72                      | 5470              | 0.268                                | 235                                        | 260                                      |
| 3 x 95                                    | 15                           | 8.0                     | -                        | 3.4                     | 77                      | 6530              | 0.193                                | 275                                        | 315                                      |
| 3 x 120                                   | 18                           | 8.0                     | -                        | 3.5                     | 80                      | 7520              | 0.153                                | 310                                        | 355                                      |
| 3 x 150                                   | 18                           | 8.0                     | -                        | 3.6                     | 83                      | 8620              | 0.124                                | 345                                        | 400                                      |
| 3 x 185                                   | 30                           | 8.0                     | -                        | 3.7                     | 87                      | 9930              | 0.0991                               | 390                                        | 455                                      |
| 3 x 240                                   | 34                           | 8.0                     | -                        | 3.9                     | 93                      | 11990             | 0.0754                               | 445                                        | 525                                      |
| 3 x 300                                   | 34                           | 8.0                     | -                        | 4.1                     | 98                      | 14160             | 0.0601                               | 490                                        | 585                                      |
| 3 x 400                                   | 53                           | 8.0                     | -                        | 4.3                     | 105                     | 17640             | 0.047                                | 540                                        | 660                                      |



## MEDIUM VOLTAGE CABLES 1.8/3(3.6) kV, XLPE INSULATED PVC SHEATHED DOUBLE ALUMINIUM TAPE ARMOUR 1 CORE CU/XLPE/CTS/PVC/DATA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Aluminium tape                            |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 1.8/3(3.6) kV                                                     |
| Reference :                       | IEC 60228 & IEC 60502-1                                           |
| AC Testing Voltage :              | 6.5kV                                                             |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Thickness of Armour Tape | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                |                                              |
| 1 x 10                                    | 6                            | 2.0                     | 2 X 0.5                  | 1.8                     | 20                      | 500               | 1.83                                 | 90                                             | 100                                          |
| 1 x 16                                    | 6                            | 2.0                     | 2 X 0.5                  | 1.8                     | 21                      | 590               | 1.15                                 | 115                                            | 125                                          |
| 1 x 25                                    | 6                            | 2.0                     | 2 X 0.5                  | 1.8                     | 22                      | 710               | 0.727                                | 150                                            | 165                                          |
| 1 x 35                                    | 6                            | 2.0                     | 2 X 0.5                  | 1.8                     | 23                      | 840               | 0.524                                | 180                                            | 200                                          |
| 1 x 50                                    | 6                            | 2.0                     | 2 X 0.5                  | 1.8                     | 25                      | 1020              | 0.387                                | 210                                            | 240                                          |
| 1 x 70                                    | 12                           | 2.0                     | 2 X 0.5                  | 1.8                     | 26                      | 1250              | 0.268                                | 260                                            | 295                                          |
| 1 x 95                                    | 15                           | 2.0                     | 2 X 0.5                  | 1.8                     | 28                      | 1540              | 0.193                                | 310                                            | 365                                          |
| 1 x 120                                   | 18                           | 2.0                     | 2 X 0.5                  | 1.9                     | 30                      | 1830              | 0.153                                | 350                                            | 420                                          |
| 1 x 150                                   | 18                           | 2.0                     | 2 X 0.5                  | 1.9                     | 31                      | 2150              | 0.124                                | 395                                            | 475                                          |
| 1 x 185                                   | 30                           | 2.0                     | 2 X 0.5                  | 2.0                     | 33                      | 2540              | 0.0991                               | 450                                            | 545                                          |
| 1 x 240                                   | 34                           | 2.0                     | 2 X 0.5                  | 2.1                     | 36                      | 3150              | 0.0754                               | 520                                            | 640                                          |
| 1 x 300                                   | 34                           | 2.0                     | 2 X 0.5                  | 2.1                     | 38                      | 3770              | 0.0601                               | 585                                            | 740                                          |
| 1 x 400                                   | 53                           | 2.0                     | 2 X 0.5                  | 2.3                     | 41                      | 4830              | 0.0470                               | 670                                            | 850                                          |

## MEDIUM VOLTAGE CABLES 3.6/6(7.2) kV, XLPE INSULATED, SHEATHED DOUBLE ALUMINIUM TAPE ARMOUR 1 CORE CU/XLPE/CTS/PVC/DATA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Aluminium tape                            |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 3.6/6(7.2) kV                                                     |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 12.5kV                                                            |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Thickness of Armour Tape | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                |                                              |
| 1 x 10                                    | 6                            | 2.5                     | 2 X 0.5                  | 1.8                     | 21                      | 540               | 1.83                                 | 90                                             | 100                                          |
| 1 x 16                                    | 6                            | 2.5                     | 2 X 0.5                  | 1.8                     | 22                      | 630               | 1.15                                 | 115                                            | 125                                          |
| 1 x 25                                    | 6                            | 2.5                     | 2 X 0.5                  | 1.8                     | 23                      | 760               | 0.727                                | 150                                            | 165                                          |
| 1 x 35                                    | 6                            | 2.5                     | 2 X 0.5                  | 1.8                     | 24                      | 890               | 0.524                                | 180                                            | 200                                          |
| 1 x 50                                    | 6                            | 2.5                     | 2 X 0.5                  | 1.8                     | 26                      | 1070              | 0.387                                | 210                                            | 240                                          |
| 1 x 70                                    | 12                           | 2.5                     | 2 X 0.5                  | 1.8                     | 27                      | 1310              | 0.268                                | 260                                            | 295                                          |
| 1 x 95                                    | 15                           | 2.5                     | 2 X 0.5                  | 1.9                     | 29                      | 1610              | 0.193                                | 310                                            | 365                                          |
| 1 x 120                                   | 18                           | 2.5                     | 2 X 0.5                  | 1.9                     | 31                      | 1890              | 0.153                                | 350                                            | 420                                          |
| 1 x 150                                   | 18                           | 2.5                     | 2 X 0.5                  | 2.0                     | 32                      | 2220              | 0.124                                | 395                                            | 475                                          |
| 1 x 185                                   | 30                           | 2.5                     | 2 X 0.5                  | 2.0                     | 34                      | 2610              | 0.0991                               | 450                                            | 545                                          |
| 1 x 240                                   | 34                           | 2.6                     | 2 X 0.5                  | 2.1                     | 37                      | 3230              | 0.0754                               | 520                                            | 640                                          |
| 1 x 300                                   | 34                           | 2.8                     | 2 X 0.5                  | 2.2                     | 40                      | 3900              | 0.0601                               | 585                                            | 740                                          |
| 1 x 400                                   | 53                           | 3.0                     | 2 X 0.5                  | 2.3                     | 44                      | 5000              | 0.0470                               | 670                                            | 850                                          |



## MEDIUM VOLTAGE CABLES 6/10(12) kV, XLPE INSULATED, PVC SHEATHED DOUBLE ALUMINIUM TAPE ARMOUR 1 CORE CU/XLPE/CTS/PVC/DATA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Aluminium tape                            |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 6/10(12) kV                                                       |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 21kV                                                              |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Thickness of Armour Tape | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                |                                              |
| 1 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 16                                    | 6                            | 3.4                     | 2 X 0.5                  | 1.8                     | 24                      | 720               | 1.15                                 | 110                                            | 120                                          |
| 1 x 25                                    | 6                            | 3.4                     | 2 X 0.5                  | 1.8                     | 25                      | 840               | 0.727                                | 140                                            | 160                                          |
| 1 x 35                                    | 6                            | 3.4                     | 2 X 0.5                  | 1.8                     | 26                      | 980               | 0.524                                | 170                                            | 195                                          |
| 1 x 50                                    | 6                            | 3.4                     | 2 X 0.5                  | 1.8                     | 28                      | 1170              | 0.387                                | 200                                            | 235                                          |
| 1 x 70                                    | 12                           | 3.4                     | 2 X 0.5                  | 1.9                     | 29                      | 1420              | 0.268                                | 250                                            | 290                                          |
| 1 x 95                                    | 15                           | 3.4                     | 2 X 0.5                  | 1.9                     | 31                      | 1710              | 0.193                                | 295                                            | 360                                          |
| 1 x 120                                   | 18                           | 3.4                     | 2 X 0.5                  | 2.0                     | 33                      | 2010              | 0.153                                | 335                                            | 415                                          |
| 1 x 150                                   | 18                           | 3.4                     | 2 X 0.5                  | 2.0                     | 34                      | 2330              | 0.124                                | 375                                            | 470                                          |
| 1 x 185                                   | 30                           | 3.4                     | 2 X 0.5                  | 2.1                     | 36                      | 2740              | 0.0991                               | 430                                            | 540                                          |
| 1 x 240                                   | 34                           | 3.4                     | 2 X 0.5                  | 2.2                     | 39                      | 3350              | 0.0754                               | 495                                            | 635                                          |
| 1 x 300                                   | 34                           | 3.4                     | 2 X 0.5                  | 2.2                     | 41                      | 3990              | 0.0601                               | 560                                            | 735                                          |
| 1 x 400                                   | 53                           | 3.4                     | 2 X 0.5                  | 2.4                     | 45                      | 5090              | 0.0470                               | 640                                            | 845                                          |

## MEDIUM VOLTAGE CABLES 8.7/15(17.5) kV, XLPE INSULATED, PVC SHEATHED DOUBLE ALUMINIUM TAPE ARMOUR 1 CORE CU/XLPE/CTS/PVC/DATA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Aluminium tape                            |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 8.7/15(17.5) kV                                                   |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 30.5kV                                                            |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Thickness of Armour Tape | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                |                                              |
| 1 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 25                                    | 6                            | 4.5                     | 2 X 0.5                  | 1.8                     | 27                      | 960               | 0.727                                | 140                                            | 155                                          |
| 1 x 35                                    | 6                            | 4.5                     | 2 X 0.5                  | 1.8                     | 29                      | 1090              | 0.524                                | 170                                            | 190                                          |
| 1 x 50                                    | 6                            | 4.5                     | 2 X 0.5                  | 1.9                     | 30                      | 1300              | 0.387                                | 200                                            | 230                                          |
| 1 x 70                                    | 12                           | 4.5                     | 2 X 0.5                  | 1.9                     | 32                      | 1540              | 0.268                                | 250                                            | 285                                          |
| 1 x 95                                    | 15                           | 4.5                     | 2 X 0.5                  | 2.0                     | 33                      | 1860              | 0.193                                | 295                                            | 355                                          |
| 1 x 120                                   | 18                           | 4.5                     | 2 X 0.5                  | 2.1                     | 35                      | 2160              | 0.153                                | 335                                            | 410                                          |
| 1 x 150                                   | 18                           | 4.5                     | 2 X 0.5                  | 2.1                     | 37                      | 2490              | 0.124                                | 375                                            | 455                                          |
| 1 x 185                                   | 30                           | 4.5                     | 2 X 0.5                  | 2.2                     | 38                      | 2900              | 0.0991                               | 430                                            | 525                                          |
| 1 x 240                                   | 34                           | 4.5                     | 2 X 0.5                  | 2.3                     | 41                      | 3530              | 0.0754                               | 495                                            | 620                                          |
| 1 x 300                                   | 34                           | 4.5                     | 2 X 0.5                  | 2.3                     | 44                      | 4190              | 0.0601                               | 560                                            | 720                                          |
| 1 x 400                                   | 53                           | 4.5                     | 2 X 0.5                  | 2.4                     | 47                      | 5270              | 0.0470                               | 640                                            | 825                                          |



## MEDIUM VOLTAGE CABLES 12/10(24) kV, XLPE INSULATED, PVC SHEATHED DOUBLE ALUMINIUM TAPE ARMOUR 1 CORE CU/XLPE/CTS/PVC/DATA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Aluminium tape                            |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 12/20(24) kV                                                      |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing                        | 42kV                                                              |
| Voltage :                         | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires (no/mm) | Thickness of insulation (mm) | Thickness of Armour Tape (mm) | Thickness of oversheath (mm) | Approx overall diameter (mm) | Approx net weight (kg/km) | Maximum Conductor Resistance at 20oC (Ohm/km) | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|--------------------------------------|------------------------------|-------------------------------|------------------------------|------------------------------|---------------------------|-----------------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 |                                      | (mm)                         | (mm)                          | (mm)                         | (mm)                         | (kg/km)                   | (Ohm/km)                                      | (A)                                            | (A)                                          |
| 1 x 10                                    | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 16                                    | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 25                                    | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 35                                    | 6                                    | 5.5                          | 2 X 0.5                       | 1.9                          | 31                           | 1220                      | 0.524                                         | 170                                            | 200                                          |
| 1 x 50                                    | 6                                    | 5.5                          | 2 X 0.5                       | 2.0                          | 32                           | 1430                      | 0.387                                         | 200                                            | 240                                          |
| 1 x 70                                    | 12                                   | 5.5                          | 2 X 0.5                       | 2.0                          | 34                           | 1680                      | 0.268                                         | 250                                            | 295                                          |
| 1 x 95                                    | 15                                   | 5.5                          | 2 X 0.5                       | 2.1                          | 36                           | 2000                      | 0.193                                         | 295                                            | 365                                          |
| 1 x 120                                   | 18                                   | 5.5                          | 2 X 0.5                       | 2.1                          | 37                           | 2300                      | 0.153                                         | 335                                            | 420                                          |
| 1 x 150                                   | 18                                   | 5.5                          | 2 X 0.5                       | 2.2                          | 39                           | 2650                      | 0.124                                         | 375                                            | 475                                          |
| 1 x 185                                   | 30                                   | 5.5                          | 2 X 0.5                       | 2.2                          | 41                           | 3050                      | 0.0991                                        | 430                                            | 545                                          |
| 1 x 240                                   | 34                                   | 5.5                          | 2 X 0.5                       | 2.3                          | 44                           | 3700                      | 0.0754                                        | 495                                            | 640                                          |
| 1 x 300                                   | 34                                   | 5.5                          | 2 X 0.5                       | 2.4                          | 46                           | 4370                      | 0.0601                                        | 560                                            | 740                                          |
| 1 x 400                                   | 53                                   | 5.5                          | 2 X 0.5                       | 2.5                          | 50                           | 5480                      | 0.0470                                        | 640                                            | 850                                          |

## MEDIUM VOLTAGE CABLES 18/30(36) kV, XLPE INSULATED, PVC SHEATHED DOUBLE ALUMINIUM TAPE ARMOUR 1 CORE CU/XLPE/CTS/PVC/DATA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Aluminium tape                            |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 18/30(36) kV                                                      |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing                        | 63kV                                                              |
| Voltage :                         | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires (no/mm) | Thickness of insulation (mm) | Thickness of Armour Tape (mm) | Thickness of oversheath (mm) | Approx overall diameter (mm) | Approx net weight (kg/km) | Maximum Conductor Resistance at 20oC (Ohm/km) | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|--------------------------------------|------------------------------|-------------------------------|------------------------------|------------------------------|---------------------------|-----------------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 |                                      | (mm)                         | (mm)                          | (mm)                         | (mm)                         | (kg/km)                   | (Ohm/km)                                      | (A)                                            | (A)                                          |
| 1 x 10                                    | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 16                                    | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 25                                    | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 35                                    | -                                    | -                            | -                             | -                            | -                            | -                         | -                                             | -                                              | -                                            |
| 1 x 50                                    | 6                                    | 8.0                          | 2 X 0.5                       | 2.1                          | 38                           | 1770                      | 0.387                                         | 200                                            | 235                                          |
| 1 x 70                                    | 12                                   | 8.0                          | 2 X 0.5                       | 2.2                          | 40                           | 2040                      | 0.268                                         | 250                                            | 290                                          |
| 1 x 95                                    | 15                                   | 8.0                          | 2 X 0.5                       | 2.3                          | 41                           | 2380                      | 0.193                                         | 295                                            | 360                                          |
| 1 x 120                                   | 18                                   | 8.0                          | 2 X 0.5                       | 2.3                          | 43                           | 2710                      | 0.153                                         | 335                                            | 415                                          |
| 1 x 150                                   | 18                                   | 8.0                          | 2 X 0.5                       | 2.4                          | 45                           | 3080                      | 0.124                                         | 375                                            | 470                                          |
| 1 x 185                                   | 30                                   | 8.0                          | 2 X 0.5                       | 2.4                          | 47                           | 3490                      | 0.0991                                        | 430                                            | 540                                          |
| 1 x 240                                   | 34                                   | 8.0                          | 2 X 0.5                       | 2.5                          | 50                           | 4170                      | 0.0754                                        | 495                                            | 635                                          |
| 1 x 300                                   | 34                                   | 8.0                          | 2 X 0.5                       | 2.6                          | 52                           | 4870                      | 0.0601                                        | 560                                            | 735                                          |
| 1 x 400                                   | 53                                   | 8.0                          | 2 X 0.5                       | 2.7                          | 56                           | 6010                      | 0.047                                         | 640                                            | 845                                          |



## MEDIUM VOLTAGE CABLES 1.8/3(3.6) kV, XLPE INSULATED, PVC SHEATHED STEEL TAPE ARMOUR 3 CORES CU/XLPE/CTS/PVC/DSTA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Galvanized steel tape                     |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 1.8/3(3.6) kV                                                     |
| Reference :                       | IEC 60228 & IEC 60502-1                                           |
| AC Testing Voltage :              | 6.5kV                                                             |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Thickness of Armour Tape | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                            | (A)                                          |
| 3 x 10                                    | 6                            | 2.0                     | 2 x 0.2                  | 1.9                     | 32                      | 1250              | 1.83                                 | 85                                             | 80                                           |
| 3 x 16                                    | 6                            | 2.0                     | 2 x 0.2                  | 2.0                     | 34                      | 1540              | 1.15                                 | 105                                            | 105                                          |
| 3 x 25                                    | 6                            | 2.0                     | 2 x 0.5                  | 2.1                     | 38                      | 2280              | 0.727                                | 135                                            | 140                                          |
| 3 x 35                                    | 6                            | 2.0                     | 2 x 0.5                  | 2.2                     | 40                      | 2710              | 0.524                                | 165                                            | 170                                          |
| 3 x 50                                    | 6                            | 2.0                     | 2 x 0.5                  | 2.3                     | 44                      | 3350              | 0.387                                | 195                                            | 205                                          |
| 3 x 70                                    | 12                           | 2.0                     | 2 x 0.5                  | 2.4                     | 47                      | 4120              | 0.268                                | 235                                            | 250                                          |
| 3 x 95                                    | 15                           | 2.0                     | 2 x 0.5                  | 2.6                     | 52                      | 5120              | 0.193                                | 280                                            | 305                                          |
| 3 x 120                                   | 18                           | 2.0                     | 2 x 0.5                  | 2.7                     | 55                      | 6080              | 0.153                                | 315                                            | 345                                          |
| 3 x 150                                   | 18                           | 2.0                     | 2 x 0.5                  | 2.8                     | 58                      | 7140              | 0.124                                | 350                                            | 390                                          |
| 3 x 185                                   | 30                           | 2.0                     | 2 x 0.5                  | 2.9                     | 63                      | 8430              | 0.0991                               | 395                                            | 445                                          |
| 3 x 240                                   | 34                           | 2.0                     | 2 x 0.5                  | 3.1                     | 69                      | 10430             | 0.0754                               | 450                                            | 520                                          |
| 3 x 300                                   | 34                           | 2.0                     | 2 x 0.5                  | 3.3                     | 75                      | 12550             | 0.0601                               | 495                                            | 580                                          |
| 3 x 400                                   | 53                           | 2.0                     | 2 x 0.8                  | 3.6                     | 83                      | 16800             | 0.0470                               | 545                                            | 655                                          |

## MEDIUM VOLTAGE CABLES 3.6/6(7.2) kV, XLPE INSULATED, PVC SHEATHED STEEL TAPE ARMOUR 3 CORES CU/XLPE/CTS/PVC/DSTA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Galvanized steel tape                     |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 3.6/6(7.2) kV                                                     |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 12.5kV                                                            |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Thickness of Armour Tape | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                            | (A)                                          |
| 3 x 10                                    | 6                            | 2.5                     | 2 x 0.2                  | 2.0                     | 34                      | 1400              | 1.83                                 | 85                                             | 80                                           |
| 3 x 16                                    | 6                            | 2.5                     | 2 x 0.2                  | 2.1                     | 37                      | 1690              | 1.15                                 | 105                                            | 105                                          |
| 3 x 25                                    | 6                            | 2.5                     | 2 x 0.5                  | 2.2                     | 41                      | 2460              | 0.727                                | 135                                            | 140                                          |
| 3 x 35                                    | 6                            | 2.5                     | 2 x 0.5                  | 2.3                     | 43                      | 2920              | 0.524                                | 165                                            | 170                                          |
| 3 x 50                                    | 6                            | 2.5                     | 2 x 0.5                  | 2.4                     | 47                      | 3550              | 0.387                                | 195                                            | 205                                          |
| 3 x 70                                    | 12                           | 2.5                     | 2 x 0.5                  | 2.5                     | 50                      | 4360              | 0.268                                | 235                                            | 250                                          |
| 3 x 95                                    | 15                           | 2.5                     | 2 x 0.5                  | 2.7                     | 54                      | 5350              | 0.193                                | 280                                            | 305                                          |
| 3 x 120                                   | 18                           | 2.5                     | 2 x 0.5                  | 2.8                     | 58                      | 6330              | 0.153                                | 315                                            | 345                                          |
| 3 x 150                                   | 18                           | 2.5                     | 2 x 0.5                  | 2.9                     | 61                      | 7420              | 0.124                                | 350                                            | 390                                          |
| 3 x 185                                   | 30                           | 2.5                     | 2 x 0.5                  | 3.0                     | 65                      | 8690              | 0.0991                               | 395                                            | 445                                          |
| 3 x 240                                   | 34                           | 2.6                     | 2 x 0.5                  | 3.2                     | 72                      | 10800             | 0.0754                               | 450                                            | 520                                          |
| 3 x 300                                   | 34                           | 2.8                     | 2 x 0.5                  | 3.4                     | 79                      | 13060             | 0.0601                               | 495                                            | 580                                          |
| 3 x 400                                   | 53                           | 3.0                     | 2 x 0.8                  | 3.7                     | 88                      | 17520             | 0.0470                               | 545                                            | 655                                          |

## MEDIUM VOLTAGE CABLES 6/10(12) kV, XLPE INSULATED, PVC SHEATHED STEEL TAPE ARMOUR 3 CORES CU/XLPE/CTS/PVC/DSTA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

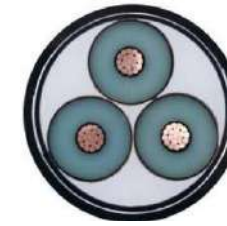
|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Galvanized steel tape                     |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 6/10(12) kV                                                       |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 21kV                                                              |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Thickness of Armour Tape | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                | (A)                                      |
| 3 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                        |
| 3 x 16                                    | 6                            | 3.4                     | 2 x 0.5                  | 2.3                     | 42                      | 2360              | 1.15                                 | 105                                            | 105                                      |
| 3 x 25                                    | 6                            | 3.4                     | 2 x 0.5                  | 2.4                     | 45                      | 2840              | 0.727                                | 135                                            | 140                                      |
| 3 x 35                                    | 6                            | 3.4                     | 2 x 0.5                  | 2.4                     | 48                      | 3270              | 0.524                                | 165                                            | 170                                      |
| 3 x 50                                    | 6                            | 3.4                     | 2 x 0.5                  | 2.6                     | 51                      | 3970              | 0.387                                | 195                                            | 210                                      |
| 3 x 70                                    | 12                           | 3.4                     | 2 x 0.5                  | 2.7                     | 55                      | 4790              | 0.268                                | 235                                            | 255                                      |
| 3 x 95                                    | 15                           | 3.4                     | 2 x 0.5                  | 2.8                     | 59                      | 5790              | 0.193                                | 280                                            | 310                                      |
| 3 x 120                                   | 18                           | 3.4                     | 2 x 0.5                  | 2.9                     | 62                      | 6790              | 0.153                                | 315                                            | 350                                      |
| 3 x 150                                   | 18                           | 3.4                     | 2 x 0.5                  | 3.0                     | 66                      | 7870              | 0.124                                | 350                                            | 395                                      |
| 3 x 185                                   | 30                           | 3.4                     | 2 x 0.5                  | 3.2                     | 71                      | 9230              | 0.0991                               | 395                                            | 450                                      |
| 3 x 240                                   | 34                           | 3.4                     | 2 x 0.5                  | 3.4                     | 77                      | 11290             | 0.0754                               | 450                                            | 525                                      |
| 3 x 300                                   | 34                           | 3.4                     | 2 x 0.8                  | 3.6                     | 83                      | 14260             | 0.0601                               | 495                                            | 585                                      |
| 3 x 400                                   | 53                           | 3.4                     | 2 x 0.8                  | 3.8                     | 91                      | 17870             | 0.0470                               | 545                                            | 660                                      |

## MEDIUM VOLTAGE CABLES 8.7/15(17.5) kV, XLPE INSULATED, PVC SHEATHED STEEL TAPE ARMOUR 3 CORES CU/XLPE/CTS/PVC/DSTA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Galvanized steel tape                     |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 8.7/15(17.5) kV                                                   |
| Reference :                       | IEC 60228 & IEC 60502-1                                           |
| AC Testing Voltage :              | 30.5kV                                                            |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Thickness of Armour Tape | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                | (A)                                      |
| 3 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                        |
| 3 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                        |
| 3 x 25                                    | 6                            | 4.5                     | 2 x 0.5                  | 2.5                     | 51                      | 3300              | 0.727                                | 135                                            | 140                                      |
| 3 x 35                                    | 6                            | 4.5                     | 2 x 0.5                  | 2.6                     | 53                      | 3780              | 0.524                                | 165                                            | 170                                      |
| 3 x 50                                    | 6                            | 4.5                     | 2 x 0.5                  | 2.7                     | 57                      | 4480              | 0.387                                | 195                                            | 210                                      |
| 3 x 70                                    | 12                           | 4.5                     | 2 x 0.5                  | 2.9                     | 60                      | 5360              | 0.268                                | 235                                            | 255                                      |
| 3 x 95                                    | 15                           | 4.5                     | 2 x 0.5                  | 3.0                     | 64                      | 6380              | 0.193                                | 280                                            | 310                                      |
| 3 x 120                                   | 18                           | 4.5                     | 2 x 0.5                  | 3.1                     | 68                      | 7410              | 0.153                                | 315                                            | 350                                      |
| 3 x 150                                   | 18                           | 4.5                     | 2 x 0.5                  | 3.2                     | 71                      | 8520              | 0.124                                | 350                                            | 395                                      |
| 3 x 185                                   | 30                           | 4.5                     | 2 x 0.5                  | 3.3                     | 76                      | 9880              | 0.0991                               | 390                                            | 450                                      |
| 3 x 240                                   | 34                           | 4.5                     | 2 x 0.8                  | 3.6                     | 84                      | 12810             | 0.0754                               | 445                                            | 525                                      |
| 3 x 300                                   | 34                           | 4.5                     | 2 x 0.8                  | 3.8                     | 89                      | 15090             | 0.0601                               | 490                                            | 585                                      |
| 3 x 400                                   | 53                           | 4.5                     | 2 x 0.8                  | 4.0                     | 97                      | 18760             | 0.0470                               | 540                                            | 660                                      |



MEDIUM VOLTAGE CABLES  
12/20(24) kV, XLPE INSULATED,  
PVC SHEATHED STEEL TAPE ARMOUR  
3 CORES CU/XLPE/CTS/PVC/DSTA/PVC



**Application**

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

**Construction:**

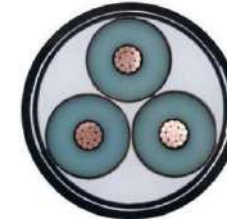
|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Galvanized steel tape                     |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

**Technical Data:**

|                                   |                                                                    |
|-----------------------------------|--------------------------------------------------------------------|
| Voltage :                         | 12/20(24) kV                                                       |
| Reference :                       | IEC 60228 & IEC 60502-1                                            |
| AC Testing                        | 42kV                                                               |
| Voltage :                         |                                                                    |
| Max. Conductor Temperature :      | 90oC (normal operation)                                            |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                          |
| Special Properties upon request : | Flame- retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Thickness of Armour Tape | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                            | (A)                                          |
| 3 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 25                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 35                                    | 6                            | 5.5                     | 2 x 0.5                  | 2.8                     | 58                      | 4280              | 0.524                                | 165                                            | 175                                          |
| 3 x 50                                    | 6                            | 5.5                     | 2 x 0.5                  | 2.9                     | 62                      | 5010              | 0.387                                | 195                                            | 215                                          |
| 3 x 70                                    | 12                           | 5.5                     | 2 x 0.5                  | 3.0                     | 65                      | 5850              | 0.268                                | 235                                            | 260                                          |
| 3 x 95                                    | 15                           | 5.5                     | 2 x 0.5                  | 3.2                     | 68                      | 6970              | 0.193                                | 280                                            | 315                                          |
| 3 x 120                                   | 18                           | 5.5                     | 2 x 0.5                  | 3.3                     | 74                      | 8020              | 0.153                                | 315                                            | 355                                          |
| 3 x 150                                   | 18                           | 5.5                     | 2 x 0.5                  | 3.4                     | 77                      | 9150              | 0.124                                | 350                                            | 400                                          |
| 3 x 185                                   | 30                           | 5.5                     | 2 x 0.5                  | 3.5                     | 81                      | 10540             | 0.0991                               | 390                                            | 455                                          |
| 3 x 240                                   | 34                           | 5.5                     | 2 x 0.8                  | 3.7                     | 88                      | 13540             | 0.0754                               | 445                                            | 525                                          |
| 3 x 300                                   | 34                           | 5.5                     | 2 x 0.8                  | 3.9                     | 94                      | 15850             | 0.0601                               | 490                                            | 585                                          |
| 3 x 400                                   | 53                           | 5.5                     | 2 x 0.8                  | 4.2                     | 101                     | 19560             | 0.0470                               | 540                                            | 660                                          |

MEDIUM VOLTAGE CABLES  
18/30(36) kV, XLPE INSULATED,  
PVC SHEATHED STEEL TAPE ARMOUR  
3 CORES CU/XLPE/CTS/PVC/DSTA/PVC



**Application**

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

**Construction:**

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Galvanized steel tape                     |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

**Technical Data:**

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 18/30(36) kV                                                      |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing                        | 63kV                                                              |
| Voltage :                         |                                                                   |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Thickness of Armour Tape | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                            | (A)                                          |
| 3 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 25                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 35                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 50                                    | 6                            | 8.0                     | 2 x 0.5                  | 3.3                     | 75                      | 6440              | 0.387                                | 195                                            | 215                                          |
| 3 x 70                                    | 12                           | 8.0                     | 2 x 0.5                  | 3.4                     | 79                      | 7380              | 0.268                                | 235                                            | 260                                          |
| 3 x 95                                    | 15                           | 8.0                     | 2 x 0.8                  | 3.6                     | 84                      | 9330              | 0.193                                | 275                                            | 315                                          |
| 3 x 120                                   | 18                           | 8.0                     | 2 x 0.8                  | 3.7                     | 88                      | 10490             | 0.153                                | 310                                            | 355                                          |
| 3 x 150                                   | 18                           | 8.0                     | 2 x 0.8                  | 3.8                     | 91                      | 11740             | 0.124                                | 345                                            | 400                                          |
| 3 x 185                                   | 30                           | 8.0                     | 2 x 0.8                  | 4.0                     | 95                      | 13240             | 0.0991                               | 390                                            | 455                                          |
| 3 x 240                                   | 34                           | 8.0                     | 2 x 0.8                  | 4.2                     | 101                     | 15560             | 0.0754                               | 445                                            | 525                                          |
| 3 x 300                                   | 34                           | 8.0                     | 2 x 0.8                  | 4.3                     | 106                     | 17930             | 0.0601                               | 490                                            | 585                                          |
| 3 x 400                                   | 53                           | 8.0                     | 2 x 0.8                  | 4.6                     | 114                     | 21820             | 0.047                                | 540                                            | 660                                          |

## MEDIUM VOLTAGE CABLES 1.8/3(3.6) kV, XLPE INSULATED, PVC SHEATHED ALUMINIUM WIRE ARMOUR 1 CORE CU/XLPE/CTS/PVC/AWA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Aluminium wire                            |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 1.8/3(3.6) kV                                                     |
| Reference :                       | IEC 60228 & IEC 60502-1                                           |
| AC Testing Voltage :              | 6.5kV                                                             |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                            | (A)                                          |
| 1 x 10                                    | 6                            | 2.0                     | 1.25                     | 1.8                     | 20                      | 540               | 1.83                                 | 90                                             | 100                                          |
| 1 x 16                                    | 6                            | 2.0                     | 1.25                     | 1.8                     | 21                      | 630               | 1.15                                 | 115                                            | 125                                          |
| 1 x 25                                    | 6                            | 2.0                     | 1.6                      | 1.8                     | 23                      | 800               | 0.727                                | 150                                            | 165                                          |
| 1 x 35                                    | 6                            | 2.0                     | 1.6                      | 1.8                     | 24                      | 940               | 0.524                                | 180                                            | 200                                          |
| 1 x 50                                    | 6                            | 2.0                     | 1.6                      | 1.8                     | 26                      | 1120              | 0.387                                | 210                                            | 240                                          |
| 1 x 70                                    | 12                           | 2.0                     | 1.6                      | 1.8                     | 27                      | 1360              | 0.268                                | 260                                            | 295                                          |
| 1 x 95                                    | 15                           | 2.0                     | 1.6                      | 1.9                     | 29                      | 1670              | 0.193                                | 310                                            | 365                                          |
| 1 x 120                                   | 18                           | 2.0                     | 1.6                      | 1.9                     | 30                      | 1950              | 0.153                                | 350                                            | 420                                          |
| 1 x 150                                   | 18                           | 2.0                     | 1.6                      | 2.0                     | 32                      | 2290              | 0.124                                | 395                                            | 475                                          |
| 1 x 185                                   | 30                           | 2.0                     | 2.0                      | 2.0                     | 34                      | 2760              | 0.0991                               | 450                                            | 545                                          |
| 1 x 240                                   | 34                           | 2.0                     | 2.0                      | 2.1                     | 37                      | 3380              | 0.0754                               | 520                                            | 640                                          |
| 1 x 300                                   | 34                           | 2.0                     | 2.0                      | 2.2                     | 40                      | 4040              | 0.0601                               | 585                                            | 740                                          |
| 1 x 400                                   | 53                           | 2.0                     | 2.0                      | 2.3                     | 43                      | 5110              | 0.0470                               | 670                                            | 850                                          |

## MEDIUM VOLTAGE CABLES 3.6/6(7.2) kV, XLPE INSULATED, PVC SHEATHED ALUMINIUM WIRE ARMOUR 1 CORE CU/XLPE/CTS/PVC/AWA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Aluminium wire                            |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 3.6/6(7.2) kV                                                     |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 12.5kV                                                            |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                            | (A)                                          |
| 1 x 10                                    | 6                            | 2.5                     | 1.25                     | 1.8                     | 22                      | 580               | 1.83                                 | 90                                             | 100                                          |
| 1 x 16                                    | 6                            | 2.5                     | 1.6                      | 1.8                     | 24                      | 720               | 1.15                                 | 115                                            | 125                                          |
| 1 x 25                                    | 6                            | 2.5                     | 1.6                      | 1.8                     | 25                      | 860               | 0.727                                | 150                                            | 165                                          |
| 1 x 35                                    | 6                            | 2.5                     | 1.6                      | 1.8                     | 26                      | 990               | 0.524                                | 180                                            | 200                                          |
| 1 x 50                                    | 6                            | 2.5                     | 1.6                      | 1.8                     | 28                      | 1180              | 0.387                                | 210                                            | 240                                          |
| 1 x 70                                    | 12                           | 2.5                     | 1.6                      | 1.8                     | 29                      | 1420              | 0.268                                | 260                                            | 295                                          |
| 1 x 95                                    | 15                           | 2.5                     | 1.6                      | 1.9                     | 31                      | 1730              | 0.193                                | 310                                            | 365                                          |
| 1 x 120                                   | 18                           | 2.5                     | 1.6                      | 2.0                     | 33                      | 2030              | 0.153                                | 350                                            | 420                                          |
| 1 x 150                                   | 18                           | 2.5                     | 2.0                      | 2.0                     | 35                      | 2440              | 0.124                                | 395                                            | 475                                          |
| 1 x 185                                   | 30                           | 2.5                     | 2.0                      | 2.1                     | 37                      | 2850              | 0.0991                               | 450                                            | 545                                          |
| 1 x 240                                   | 34                           | 2.6                     | 2.0                      | 2.2                     | 40                      | 3490              | 0.0754                               | 520                                            | 640                                          |
| 1 x 300                                   | 34                           | 2.8                     | 2.0                      | 2.3                     | 43                      | 4190              | 0.0601                               | 585                                            | 740                                          |
| 1 x 400                                   | 53                           | 3.0                     | 2.5                      | 2.4                     | 47                      | 5450              | 0.0470                               | 670                                            | 850                                          |



## MEDIUM VOLTAGE CABLES 6/10(12) kV, XLPE INSULATED, PVC SHEATHED ALUMINIUM WIRE ARMOUR 1 CORE CU/XLPE/CTS/PVC/AWA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Aluminium wire                            |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 6/10(12) kV                                                       |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 21kV                                                              |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                            | (A)                                          |
| 1 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 16                                    | 6                            | 3.4                     | 1.6                      | 1.8                     | 25                      | 810               | 1.15                                 | 110                                            | 120                                          |
| 1 x 25                                    | 6                            | 3.4                     | 1.6                      | 1.8                     | 26                      | 940               | 0.727                                | 140                                            | 160                                          |
| 1 x 35                                    | 6                            | 3.4                     | 1.6                      | 1.8                     | 27                      | 1080              | 0.524                                | 170                                            | 195                                          |
| 1 x 50                                    | 6                            | 3.4                     | 1.6                      | 1.9                     | 29                      | 1290              | 0.387                                | 200                                            | 235                                          |
| 1 x 70                                    | 12                           | 3.4                     | 1.6                      | 1.9                     | 30                      | 1530              | 0.268                                | 250                                            | 290                                          |
| 1 x 95                                    | 15                           | 3.4                     | 1.6                      | 2.0                     | 32                      | 1850              | 0.193                                | 295                                            | 360                                          |
| 1 x 120                                   | 18                           | 3.4                     | 2.0                      | 2.0                     | 35                      | 2230              | 0.153                                | 335                                            | 415                                          |
| 1 x 150                                   | 18                           | 3.4                     | 2.0                      | 2.1                     | 36                      | 2580              | 0.124                                | 375                                            | 470                                          |
| 1 x 185                                   | 30                           | 3.4                     | 2.0                      | 2.2                     | 38                      | 2990              | 0.0991                               | 430                                            | 540                                          |
| 1 x 240                                   | 34                           | 3.4                     | 2.0                      | 2.2                     | 41                      | 3610              | 0.0754                               | 495                                            | 635                                          |
| 1 x 300                                   | 34                           | 3.4                     | 2.0                      | 2.3                     | 43                      | 4270              | 0.0601                               | 560                                            | 735                                          |
| 1 x 400                                   | 53                           | 3.4                     | 2.5                      | 2.5                     | 48                      | 5540              | 0.0470                               | 640                                            | 845                                          |

## MEDIUM VOLTAGE CABLES 8.7/15(17.5) kV, XLPE INSULATED, PVC SHEATHED ALUMINIUM WIRE ARMOUR 1 CORE CU/XLPE/CTS/PVC/AWA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Aluminium wire                            |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 8.7/15(17.5) kV                                                   |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 30.5kV                                                            |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                            | (A)                                          |
| 1 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 25                                    | 6                            | 4.5                     | 1.6                      | 1.8                     | 28                      | 1070              | 0.727                                | 140                                            | 155                                          |
| 1 x 35                                    | 6                            | 4.5                     | 1.6                      | 1.9                     | 30                      | 1220              | 0.524                                | 170                                            | 190                                          |
| 1 x 50                                    | 6                            | 4.5                     | 1.6                      | 1.9                     | 31                      | 1425              | 0.387                                | 200                                            | 230                                          |
| 1 x 70                                    | 12                           | 4.5                     | 1.6                      | 2.0                     | 33                      | 1690              | 0.268                                | 250                                            | 285                                          |
| 1 x 95                                    | 15                           | 4.5                     | 2.0                      | 2.1                     | 36                      | 2095              | 0.193                                | 295                                            | 355                                          |
| 1 x 120                                   | 18                           | 4.5                     | 2.0                      | 2.1                     | 37                      | 2395              | 0.153                                | 335                                            | 410                                          |
| 1 x 150                                   | 18                           | 4.5                     | 2.0                      | 2.2                     | 39                      | 2755              | 0.124                                | 375                                            | 455                                          |
| 1 x 185                                   | 30                           | 4.5                     | 2.0                      | 2.2                     | 41                      | 3160              | 0.0991                               | 430                                            | 525                                          |
| 1 x 240                                   | 34                           | 4.5                     | 2.0                      | 2.3                     | 43                      | 3810              | 0.0754                               | 495                                            | 620                                          |
| 1 x 300                                   | 34                           | 4.5                     | 2.5                      | 2.4                     | 47                      | 4640              | 0.0601                               | 560                                            | 720                                          |
| 1 x 400                                   | 53                           | 4.5                     | 2.5                      | 2.5                     | 50                      | 5755              | 0.0470                               | 640                                            | 825                                          |

MEDIUM VOLTAGE CABLES  
12/20(24) kV, XLPE INSULATED,  
PVC SHEATHED ALUMINIUM WIRE ARMOUR  
1 CORE CU/XLPE/CTS/PVC/AWA/PVC



**Application**

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

**Construction:**

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Aluminium wire                            |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

**Technical Data:**

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 12/20(24) kV                                                      |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 42kV                                                              |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                            | (A)                                          |
| 1 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 25                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 35                                    | 6                            | 5.5                     | 1.6                      | 2.0                     | 32                      | 1360              | 0.524                                | 170                                            | 200                                          |
| 1 x 50                                    | 6                            | 5.5                     | 2.0                      | 2.0                     | 34                      | 1645              | 0.387                                | 200                                            | 240                                          |
| 1 x 70                                    | 12                           | 5.5                     | 2.0                      | 2.1                     | 36                      | 1925              | 0.268                                | 250                                            | 295                                          |
| 1 x 95                                    | 15                           | 5.5                     | 2.0                      | 2.1                     | 38                      | 2240              | 0.193                                | 295                                            | 365                                          |
| 1 x 120                                   | 18                           | 5.5                     | 2.0                      | 2.2                     | 39                      | 2560              | 0.153                                | 335                                            | 420                                          |
| 1 x 150                                   | 18                           | 5.5                     | 2.0                      | 2.2                     | 41                      | 2905              | 0.124                                | 375                                            | 475                                          |
| 1 x 185                                   | 30                           | 5.5                     | 2.0                      | 2.3                     | 43                      | 3340              | 0.0991                               | 430                                            | 545                                          |
| 1 x 240                                   | 34                           | 5.5                     | 2.5                      | 2.4                     | 47                      | 4150              | 0.0754                               | 495                                            | 640                                          |
| 1 x 300                                   | 34                           | 5.5                     | 2.5                      | 2.5                     | 49                      | 4850              | 0.0601                               | 560                                            | 740                                          |
| 1 x 400                                   | 53                           | 5.5                     | 2.5                      | 2.6                     | 53                      | 6000              | 0.0470                               | 640                                            | 850                                          |

MEDIUM VOLTAGE CABLES  
18/30(36) kV, XLPE INSULATED,  
PVC SHEATHED ALUMINIUM WIRE ARMOUR  
1 CORE CU/XLPE/CTS/PVC/AWA/PVC



**Application**

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

**Construction:**

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Aluminium wire                            |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

**Technical Data:**

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 18/30(36) kV                                                      |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 63kV                                                              |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                            | (A)                                          |
| 1 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 25                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 35                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 1 x 50                                    | 6                            | 8.0                     | 2.0                      | 2.2                     | 40                      | 2030              | 0.387                                | 200                                            | 235                                          |
| 1 x 70                                    | 12                           | 8.0                     | 2.0                      | 2.3                     | 42                      | 2320              | 0.268                                | 250                                            | 290                                          |
| 1 x 95                                    | 15                           | 8.0                     | 2.0                      | 2.3                     | 43                      | 2660              | 0.193                                | 295                                            | 360                                          |
| 1 x 120                                   | 18                           | 8.0                     | 2.5                      | 2.4                     | 46                      | 3160              | 0.153                                | 335                                            | 415                                          |
| 1 x 150                                   | 18                           | 8.0                     | 2.5                      | 2.5                     | 48                      | 3530              | 0.124                                | 375                                            | 470                                          |
| 1 x 185                                   | 30                           | 8.0                     | 2.5                      | 2.5                     | 50                      | 3980              | 0.0991                               | 430                                            | 540                                          |
| 1 x 240                                   | 34                           | 8.0                     | 2.5                      | 2.6                     | 53                      | 4690              | 0.0754                               | 495                                            | 635                                          |
| 1 x 300                                   | 34                           | 8.0                     | 2.5                      | 2.7                     | 56                      | 5400              | 0.0601                               | 560                                            | 735                                          |
| 1 x 400                                   | 53                           | 8.0                     | 2.5                      | 2.8                     | 59                      | 6580              | 0.047                                | 640                                            | 845                                          |



## MEDIUM VOLTAGE CABLES 1.8/3(3.6) kV, XLPE INSULATED, PVC SHEATHED STEEL WIRE ARMOUR 3 CORES CU/XLPE/CTS/PVC/SWA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

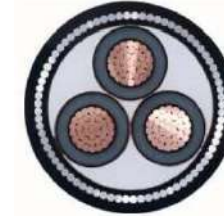
|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Galvanized steel wire                     |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 1.8/3(3.6) kV                                                     |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 6.5Kv                                                             |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                |                                              |
| 3 x 10                                    | 6                            | 2.0                     | 2.0                      | 2.0                     | 35                      | 2160              | 1.83                                 | 85                                             | 80                                           |
| 3 x 16                                    | 6                            | 2.0                     | 2.0                      | 2.1                     | 38                      | 2500              | 1.15                                 | 105                                            | 105                                          |
| 3 x 25                                    | 6                            | 2.0                     | 2.0                      | 2.2                     | 40                      | 2980              | 0.727                                | 135                                            | 140                                          |
| 3 x 35                                    | 6                            | 2.0                     | 2.0                      | 2.3                     | 43                      | 3450              | 0.524                                | 165                                            | 170                                          |
| 3 x 50                                    | 6                            | 2.0                     | 2.5                      | 2.4                     | 47                      | 4540              | 0.387                                | 195                                            | 205                                          |
| 3 x 70                                    | 12                           | 2.0                     | 2.5                      | 2.5                     | 51                      | 5420              | 0.268                                | 235                                            | 250                                          |
| 3 x 95                                    | 15                           | 2.0                     | 2.5                      | 2.7                     | 55                      | 6560              | 0.193                                | 280                                            | 305                                          |
| 3 x 120                                   | 18                           | 2.0                     | 2.5                      | 2.8                     | 59                      | 7620              | 0.153                                | 315                                            | 345                                          |
| 3 x 150                                   | 18                           | 2.0                     | 2.5                      | 2.9                     | 62                      | 8740              | 0.124                                | 350                                            | 390                                          |
| 3 x 185                                   | 30                           | 2.0                     | 2.5                      | 3.0                     | 66                      | 10160             | 0.0991                               | 395                                            | 445                                          |
| 3 x 240                                   | 34                           | 2.0                     | 2.5                      | 3.2                     | 72                      | 12350             | 0.0754                               | 450                                            | 520                                          |
| 3 x 300                                   | 34                           | 2.0                     | 3.15                     | 3.5                     | 80                      | 15520             | 0.0601                               | 495                                            | 580                                          |
| 3 x 400                                   | 53                           | 2.0                     | 3.15                     | 3.7                     | 87                      | 19220             | 0.0470                               | 545                                            | 655                                          |

## MEDIUM VOLTAGE CABLES 3.6/6(7.2) kV, XLPE INSULATED, PVC SHEATHED STEEL WIRE ARMOUR 3 CORES CU/XLPE/CTS/PVC/SWA/PVC



### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Galvanized steel wire                     |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 3.6/6(7.2) kV                                                     |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 12.5kV                                                            |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                |                                              |
| 3 x 10                                    | 6                            | 2.5                     | 2.0                      | 2.1                     | 38                      | 2360              | 1.83                                 | 85                                             | 80                                           |
| 3 x 16                                    | 6                            | 2.5                     | 2.0                      | 2.2                     | 40                      | 2720              | 1.15                                 | 105                                            | 105                                          |
| 3 x 25                                    | 6                            | 2.5                     | 2.0                      | 2.3                     | 43                      | 3200              | 0.727                                | 135                                            | 140                                          |
| 3 x 35                                    | 6                            | 2.5                     | 2.5                      | 2.4                     | 47                      | 4090              | 0.524                                | 165                                            | 170                                          |
| 3 x 50                                    | 6                            | 2.5                     | 2.5                      | 2.5                     | 50                      | 4830              | 0.387                                | 195                                            | 205                                          |
| 3 x 70                                    | 12                           | 2.5                     | 2.5                      | 2.6                     | 53                      | 5740              | 0.268                                | 235                                            | 250                                          |
| 3 x 95                                    | 15                           | 2.5                     | 2.5                      | 2.8                     | 57                      | 6830              | 0.193                                | 280                                            | 305                                          |
| 3 x 120                                   | 18                           | 2.5                     | 2.5                      | 2.9                     | 61                      | 7900              | 0.153                                | 315                                            | 345                                          |
| 3 x 150                                   | 18                           | 2.5                     | 2.5                      | 3.0                     | 64                      | 9100              | 0.124                                | 350                                            | 390                                          |
| 3 x 185                                   | 30                           | 2.5                     | 2.5                      | 3.1                     | 69                      | 10510             | 0.0991                               | 395                                            | 445                                          |
| 3 x 240                                   | 34                           | 2.6                     | 3.15                     | 3.4                     | 77                      | 13680             | 0.0754                               | 450                                            | 520                                          |
| 3 x 300                                   | 34                           | 2.8                     | 3.15                     | 3.6                     | 84                      | 16160             | 0.0601                               | 495                                            | 580                                          |
| 3 x 400                                   | 53                           | 3.0                     | 3.15                     | 3.9                     | 92                      | 20160             | 0.0470                               | 545                                            | 655                                          |

## MEDIUM VOLTAGE CABLES

### 6/10(12) kV, XLPE INSULATED, PVC SHEATHED STEEL WIRE ARMOUR 3 CORES CU/XLPE/CTS/PVC/SWA/PVC



#### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

#### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Galvanized steel wire                     |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

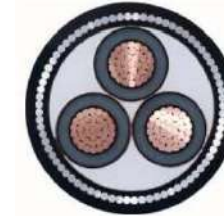
#### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 6/10(12) kV                                                       |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 21kV                                                              |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC | Ampacities in free air at 40oC (ambient) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|--------------------------------------------|------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                        | (A)                                      |
| 3 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                          | -                                        |
| 3 x 16                                    | 6                            | 3.4                     | 2.0                      | 2.3                     | 44                      | 3120              | 1.15                                 | 105                                        | 105                                      |
| 3 x 25                                    | 6                            | 3.4                     | 2.5                      | 2.5                     | 49                      | 4090              | 0.727                                | 135                                        | 140                                      |
| 3 x 35                                    | 6                            | 3.4                     | 2.5                      | 2.6                     | 50                      | 4590              | 0.524                                | 165                                        | 170                                      |
| 3 x 50                                    | 6                            | 3.4                     | 2.5                      | 2.7                     | 54                      | 5370              | 0.387                                | 195                                        | 210                                      |
| 3 x 70                                    | 12                           | 3.4                     | 2.5                      | 2.8                     | 58                      | 6300              | 0.268                                | 235                                        | 255                                      |
| 3 x 95                                    | 15                           | 3.4                     | 2.5                      | 2.9                     | 62                      | 7390              | 0.193                                | 280                                        | 310                                      |
| 3 x 120                                   | 18                           | 3.4                     | 2.5                      | 3.0                     | 66                      | 8530              | 0.153                                | 315                                        | 350                                      |
| 3 x 150                                   | 18                           | 3.4                     | 2.5                      | 3.1                     | 69                      | 9680              | 0.124                                | 350                                        | 395                                      |
| 3 x 185                                   | 30                           | 3.4                     | 2.5                      | 3.3                     | 73                      | 11180             | 0.0991                               | 395                                        | 450                                      |
| 3 x 240                                   | 34                           | 3.4                     | 3.15                     | 3.5                     | 81                      | 14270             | 0.0754                               | 450                                        | 525                                      |
| 3 x 300                                   | 34                           | 3.4                     | 3.15                     | 3.7                     | 87                      | 16680             | 0.0601                               | 495                                        | 585                                      |
| 3 x 400                                   | 53                           | 3.4                     | 3.15                     | 3.9                     | 94                      | 20540             | 0.0470                               | 545                                        | 660                                      |

## MEDIUM VOLTAGE CABLES

### 8.7/15(17.5) kV, XLPE INSULATED, PVC SHEATHED STEEL WIRE ARMOUR 3 CORES CU/XLPE/CTS/PVC/SWA/PVC



#### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

#### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Galvanized steel wire                     |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

#### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 8.7/15(17.5) kV                                                   |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 30.5kV                                                            |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC | Ampacities in free air at 40oC (ambient) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|--------------------------------------------|------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                        | (A)                                      |
| 3 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                          | -                                        |
| 3 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                          | -                                        |
| 3 x 25                                    | 6                            | 4.5                     | 2.5                      | 2.7                     | 54                      | 4730              | 0.727                                | 135                                        | 140                                      |
| 3 x 35                                    | 6                            | 4.5                     | 2.5                      | 2.7                     | 56                      | 5220              | 0.524                                | 165                                        | 170                                      |
| 3 x 50                                    | 6                            | 4.5                     | 2.5                      | 2.8                     | 60                      | 6030              | 0.387                                | 195                                        | 210                                      |
| 3 x 70                                    | 12                           | 4.5                     | 2.5                      | 3.0                     | 64                      | 7020              | 0.268                                | 235                                        | 255                                      |
| 3 x 95                                    | 15                           | 4.5                     | 2.5                      | 3.1                     | 68                      | 8180              | 0.193                                | 280                                        | 310                                      |
| 3 x 120                                   | 18                           | 4.5                     | 2.5                      | 3.2                     | 72                      | 9310              | 0.153                                | 315                                        | 350                                      |
| 3 x 150                                   | 18                           | 4.5                     | 3.15                     | 3.4                     | 77                      | 11340             | 0.124                                | 350                                        | 395                                      |
| 3 x 185                                   | 30                           | 4.5                     | 3.15                     | 3.5                     | 81                      | 12890             | 0.0991                               | 390                                        | 450                                      |
| 3 x 240                                   | 34                           | 4.5                     | 3.15                     | 3.7                     | 87                      | 15220             | 0.0754                               | 445                                        | 525                                      |
| 3 x 300                                   | 34                           | 4.5                     | 3.15                     | 3.9                     | 92                      | 17680             | 0.0601                               | 490                                        | 585                                      |
| 3 x 400                                   | 53                           | 4.5                     | 3.15                     | 4.1                     | 100                     | 21540             | 0.0470                               | 540                                        | 660                                      |



## MEDIUM VOLTAGE CABLES

### 12/20(24) kV, XLPE INSULATED, PVC SHEATHED STEEL WIRE ARMOUR 3 CORES CU/XLPE/CTS/PVC/SWA/PVC



#### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

#### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Galvanized steel wire                     |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

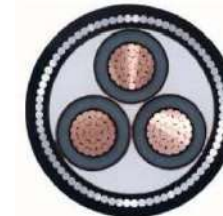
#### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 12/20(24) kV                                                      |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 42kV                                                              |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                |                                              |
| 3 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 25                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 35                                    | 6                            | 5.5                     | 2.5                      | 2.9                     | 62                      | 5980              | 0.524                                | 165                                            | 175                                          |
| 3 x 50                                    | 6                            | 5.5                     | 2.5                      | 3.0                     | 65                      | 6720              | 0.387                                | 195                                            | 215                                          |
| 3 x 70                                    | 12                           | 5.5                     | 2.5                      | 3.1                     | 69                      | 7670              | 0.268                                | 235                                            | 260                                          |
| 3 x 95                                    | 15                           | 5.5                     | 2.5                      | 3.3                     | 73                      | 8880              | 0.193                                | 280                                            | 315                                          |
| 3 x 120                                   | 18                           | 5.5                     | 3.15                     | 3.4                     | 79                      | 10910             | 0.153                                | 315                                            | 355                                          |
| 3 x 150                                   | 18                           | 5.5                     | 3.15                     | 3.5                     | 82                      | 12180             | 0.124                                | 350                                            | 400                                          |
| 3 x 185                                   | 30                           | 5.5                     | 3.15                     | 3.7                     | 86                      | 13730             | 0.0991                               | 390                                            | 455                                          |
| 3 x 240                                   | 34                           | 5.5                     | 3.15                     | 3.9                     | 92                      | 16170             | 0.0754                               | 445                                            | 525                                          |
| 3 x 300                                   | 34                           | 5.5                     | 3.15                     | 4.0                     | 97                      | 18560             | 0.0601                               | 490                                            | 585                                          |
| 3 x 400                                   | 53                           | 5.5                     | 3.15                     | 4.2                     | 105                     | 22530             | 0.0470                               | 540                                            | 660                                          |

## MEDIUM VOLTAGE CABLES

### 18/30(36) kV, XLPE INSULATED, PVC SHEATHED STEEL WIRE ARMOUR 3 CORES CU/XLPE/CTS/PVC/SWA/PVC



#### Application

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

#### Construction:

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      | Black PVC (Polyvinyl Chloride)            |
| Armour :            | Galvanized steel wire                     |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

#### Technical Data:

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 18/30(36) kV                                                      |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 63kV                                                              |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) (A) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             |                                                |                                              |
| 3 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 25                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 35                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                            |
| 3 x 50                                    | 6                            | 8.0                     | 3.15                     | 3.5                     | 80                      | 9400              | 0.387                                | 195                                            | 215                                          |
| 3 x 70                                    | 12                           | 8.0                     | 3.15                     | 3.6                     | 83                      | 10480             | 0.268                                | 235                                            | 260                                          |
| 3 x 95                                    | 15                           | 8.0                     | 3.15                     | 3.7                     | 87                      | 11790             | 0.193                                | 275                                            | 315                                          |
| 3 x 120                                   | 18                           | 8.0                     | 3.15                     | 3.8                     | 91                      | 13040             | 0.153                                | 310                                            | 355                                          |
| 3 x 150                                   | 18                           | 8.0                     | 3.15                     | 3.9                     | 95                      | 14400             | 0.124                                | 345                                            | 400                                          |
| 3 x 185                                   | 30                           | 8.0                     | 3.15                     | 4.1                     | 98                      | 15990             | 0.0991                               | 390                                            | 455                                          |
| 3 x 240                                   | 34                           | 8.0                     | 3.15                     | 4.3                     | 105                     | 18540             | 0.0754                               | 445                                            | 525                                          |
| 3 x 300                                   | 34                           | 8.0                     | 3.15                     | 4.4                     | 110                     | 21080             | 0.0601                               | 490                                            | 585                                          |
| 3 x 400                                   | 53                           | 8.0                     | 3.15                     | 4.7                     | 117                     | 25160             | 0.047                                | 540                                            | 660                                          |

MEDIUM VOLTAGE CABLES  
15 kV, 133% XLPE INSULATED,  
PVC SHEATHED NON-ARMOUR  
1 CORE CU/XLPE/CTS/PVC



**Application**

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

**Construction:**

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape - -            |
| Inner Sheath :      | Polyester / Non-woven tape                |
| Armour :            | Black PVC (Polyvinyl Chloride)            |
| Wrapping Tape :     |                                           |
| Outer Sheath :      |                                           |

**Technical Data:**

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 15 kV                                                             |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 33kV                                                              |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                            | (A)                                      |
| 1 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                        |
| 1 x 16                                    | 6                            | 5.46                    | -                        | 2.0                     | 26                      | 790               | 1.15                                 | 140                                            | 155                                      |
| 1 x 25                                    | 6                            | 5.46                    | -                        | 2.0                     | 27                      | 920               | 0.727                                | 170                                            | 190                                      |
| 1 x 35                                    | 6                            | 5.46                    | -                        | 2.0                     | 28                      | 1100              | 0.524                                | 200                                            | 230                                      |
| 1 x 50                                    | 6                            | 5.46                    | -                        | 2.0                     | 30                      | 1330              | 0.387                                | 250                                            | 285                                      |
| 1 x 70                                    | 12                           | 5.46                    | -                        | 2.0                     | 32                      | 1620              | 0.268                                | 295                                            | 355                                      |
| 1 x 95                                    | 15                           | 5.46                    | -                        | 2.0                     | 33                      | 1900              | 0.193                                | 335                                            | 410                                      |
| 1 x 120                                   | 18                           | 5.46                    | -                        | 2.0                     | 35                      | 2220              | 0.153                                | 375                                            | 455                                      |
| 1 x 150                                   | 18                           | 5.46                    | -                        | 2.0                     | 37                      | 2600              | 0.124                                | 430                                            | 525                                      |
| 1 x 185                                   | 30                           | 5.46                    | -                        | 2.0                     | 39                      | 3180              | 0.0991                               | 495                                            | 620                                      |
| 1 x 240                                   | 34                           | 5.46                    | -                        | 2.0                     | 42                      | 3800              | 0.0754                               | 650                                            | 720                                      |
| 1 x 300                                   | 34                           | 5.46                    | -                        | 2.0                     | 46                      | 4980              | 0.0601                               | 640                                            | 825                                      |
| 1 x 400                                   | 53                           | 5.46                    | -                        | 2.7                     | 50                      | 6020              | 0.0470                               | 725                                            | 950                                      |

MEDIUM VOLTAGE CABLES  
25 kV, 133% XLPE INSULATED,  
PVC SHEATHED NON-ARMOUR  
1 CORE CU/XLPE/CTS/PVC



**Application**

Used for urban networks. Exposed in aerial, direct burial, conduit, open tray and underground duct installation.

**Construction:**

|                     |                                           |
|---------------------|-------------------------------------------|
| Conductor :         | Compact round stranded anneal copper      |
| Conductor Screen :  | Semi-conducting Cross-linked Polyethylene |
| Insulation :        | XLPE (Cross-linked Polyethylene)          |
| Insulation Screen : | Semi-conducting Cross-linked Polyethylene |
| Metallic Shield :   | Annealed copper tape (CTS)                |
| Filler :            | Non-hygroscopic material                  |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Inner Sheath :      |                                           |
| Armour :            |                                           |
| Wrapping Tape :     | Polyester / Non-woven tape                |
| Outer Sheath :      | Black PVC (Polyvinyl Chloride)            |

**Technical Data:**

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Voltage :                         | 25 kV                                                             |
| Reference :                       | IEC 60228 & IEC 60502-2                                           |
| AC Testing Voltage :              | 49kV                                                              |
| Max. Conductor Temperature :      | 90oC (normal operation)                                           |
| Max. Conductor Temperature :      | 250oC (short circuit at 5s max. duration)                         |
| Special Properties upon request : | Flame-retardant sheath<br>LSHF sheath<br>(PE) Polyethylene sheath |

| Nominal cross-sectional area of conductor | Number and diameter of wires | Thickness of insulation | Nom. Dia. Of Armour Wire | Thickness of oversheath | Approx overall diameter | Approx net weight | Maximum Conductor Resistance at 20oC | Ampacities direct burial in ground at 30oC (A) | Ampacities in free air at 40oC (ambient) |
|-------------------------------------------|------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------|--------------------------------------|------------------------------------------------|------------------------------------------|
| Core x (mm <sup>2</sup> )                 | (no/mm)                      | (mm)                    | (mm)                     | (mm)                    | (mm)                    | (kg/km)           | (Ohm/km)                             | (A)                                            | (A)                                      |
| 1 x 10                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                        |
| 1 x 16                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                        |
| 1 x 25                                    | -                            | -                       | -                        | -                       | -                       | -                 | -                                    | -                                              | -                                        |
| 1 x 35                                    | 6                            | 8.76                    | -                        | 2.0                     | 34                      | 1260              | 0.524                                | 170                                            | 200                                      |
| 1 x 50                                    | 6                            | 8.76                    | -                        | 2.0                     | 36                      | 1460              | 0.387                                | 200                                            | 240                                      |
| 1 x 70                                    | 12                           | 8.76                    | -                        | 2.0                     | 37                      | 1710              | 0.268                                | 250                                            | 295                                      |
| 1 x 95                                    | 15                           | 8.76                    | -                        | 2.0                     | 39                      | 2010              | 0.193                                | 295                                            | 365                                      |
| 1 x 120                                   | 18                           | 8.76                    | -                        | 2.0                     | 41                      | 2300              | 0.153                                | 335                                            | 420                                      |
| 1 x 150                                   | 18                           | 8.76                    | -                        | 2.0                     | 42                      | 2640              | 0.124                                | 375                                            | 475                                      |
| 1 x 185                                   | 30                           | 8.76                    | -                        | 2.7                     | 45                      | 3180              | 0.0991                               | 430                                            | 545                                      |
| 1 x 240                                   | 34                           | 8.76                    | -                        | 2.7                     | 48                      | 3800              | 0.0754                               | 495                                            | 640                                      |
| 1 x 300                                   | 34                           | 8.76                    | -                        | 2.7                     | 50                      | 4460              | 0.0601                               | 560                                            | 740                                      |
| 1 x 400                                   | 53                           | 8.76                    | -                        | 2.7                     | 53                      | 5520              | 0.0470                               | 640                                            | 850                                      |





HONGKONG Kai Tak Sports Park

HONGKONG YOHO-WEST



HONGKONG NOVO Land



Nanning International Convention and Exhibition Center







Shiji Bridge



Nur Sultan City Government



OLYMPIC STADIUM



Pazhou International Convention and Exhibition Center



STAR CITY



ONE PARK



# Global Layout

## Asia



## Africa



## Europe



## North America



## South America



## Oceania



POWERING  
THE FUTURE